

# SUPPLEMENTARY NOTES — EXAM-FOCUSED

## Preparation of Patient for Operation • Pre & Post Operative Patient Care

Only the gaps in the studied PDF (Ch. 1 Perioperative Care of the Patient • Ch. 2 Protection of the Patient in Surgery • Ch. 3 Safety Measures for OR Personnel), arranged in syllabus order

### SCOPE RULE USED FOR THIS DOCUMENT

**Sections are numbered to match your syllabus sub-points exactly**, so you can tick them off one by one.

**Kept:** every gap that sits under a listed sub-point — admission procedure, transfer/position, environmental controls, electrosurgery, operative records, counting, **sterilization**, emergencies & disasters, in-service education, body mechanics, fatigue, radiation safety, infection control, chemical hazards, plus **post-operative care** (named in the unit heading).

**Cut as off-syllabus:** ASA/Mallampati and cardiac-risk scoring, the routine-investigation list, premedication pharmacology, peri-operative anticoagulant/herbal management, ERAS, post-operative fluid and nutrition regimens, analgesic pharmacology in depth, discharge medication reconciliation, laser classes, local-anaesthetic toxicity, and the single-use-device reuse debate. These belong to anaesthesia, clinical-pharmacy or equipment units, not to this one.

**Revise from the PDF, not from here:** psychological support (Ch. 1 is excellent), the record forms, the position figures, the 9 counting rules, orientation and fatigue text. This document only adds what those pages leave out.

## 1. PRE-OPERATIVE CONSIDERATIONS — PSYCHOLOGICAL SUPPORT

Your PDF covers this sub-point thoroughly. Only four things are missing, and all four are one-line answers.

| Point                            | Content                                                                                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sources of pre-operative anxiety | Fear of the <b>unknown</b> , of <b>death</b> , of <b>not waking from anaesthesia</b> , of <b>pain</b> , of <b>disfigurement/altered body image</b> , of <b>loss of control and dependence</b> ; plus separation from family and financial worry |
| Assessment tools                 | <b>STAI</b> (State-Trait Anxiety Inventory) • <b>APAIS</b> (Amsterdam Preoperative Anxiety and Information Scale) • Visual Analogue Anxiety Scale • <b>m-YPAS</b> for children                                                                  |
| Peak anxiety                     | <b>Immediately before transfer to the OR</b> (also on being told surgery is needed, and the night before)                                                                                                                                       |
| Effects of unrelieved anxiety    | ↑ catecholamines → tachycardia, hypertension, arrhythmia • ↑ anaesthetic requirement • ↑ <b>post-operative pain scores and analgesic need</b> • ↑ PONV • delayed wound healing • longer stay                                                    |
| Most effective intervention      | <b>Structured pre-operative teaching/counselling</b> . Then relaxation and guided imagery, music, parental presence at induction in children, allowing spectacles/hearing aids/dentures to stay until induction                                 |
| Age-specific fears               | <b>1-3 y</b> separation • <b>3-6 y</b> mutilation • <b>school age</b> loss of control • <b>elderly</b> dependency and death (needs slower, repeated explanation)                                                                                |

## 2. PREPARATION OF THE PATIENT FOR OPERATION (PHYSICAL PREPARATION)

The PDF reproduces the pre-operative checklist form but never explains the preparation itself.

### 2.1 Pre-operative fasting — remember 2 - 4 - 6 - 8

| Ingested material                                        | Minimum fast   |
|----------------------------------------------------------|----------------|
| Clear liquids (water, pulp-free juice, black tea/coffee) | <b>2 hours</b> |
| Breast milk                                              | <b>4 hours</b> |
| Infant formula, non-human milk, light meal               | <b>6 hours</b> |
| Fried/fatty food, meat, full meal                        | <b>8 hours</b> |

#### WHY — and when the rule does not apply

Purpose = prevent **pulmonary aspiration of gastric contents (Mendelson's syndrome)** at induction, when airway reflexes are abolished. Risk rises when gastric volume is **> 25 mL (0.4 mL/kg)** and **pH < 2.5**.

**Treat as a full stomach regardless of fasting time:** emergency and trauma, pregnancy, obesity, diabetic gastroparesis, bowel obstruction, severe GERD, opioids → **rapid sequence induction with cricoid pressure (Sellick's manoeuvre)** and acid prophylaxis.

**Prolonged 'midnight' fasting is harmful** (dehydration, hypoglycaemia, insulin resistance) and does **not** reduce gastric volume.

## 2.2 Skin preparation and hair removal — evidence the PDF predates

| Question                | Answer                                                                                                                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Should hair be removed? | <b>No</b> — not unless it physically interferes with the procedure or closure                                                                                                                    |
| If it must be removed?  | <b>Electric clippers with a single-use head</b> , or depilatory cream. <b>Razors are contraindicated</b> (micro-abrasions become colonised)                                                      |
| When?                   | <b>Immediately before surgery, outside the OR.</b> SSI risk: no removal ≈ clipping < shaving immediately before < <b>shaving the night before (highest)</b>                                      |
| Best antiseptic         | <b>Alcohol-based 2 % chlorhexidine gluconate</b> > aqueous povidone-iodine. Povidone-iodine 10 % = 1 % available iodine and <b>needs 2-3 min contact and must dry</b>                            |
| Alcohol + ESU           | Prep must <b>dry completely (≥ 3 min) and must never pool</b> — pooled alcohol is the classic <b>fuel</b> in OR fires (§10)                                                                      |
| Direction               | From the <b>incision outward</b> in concentric strokes; the sponge is <b>never returned to the centre</b> ; the dirtiest area (stoma, sinus, perineum, umbilicus, axilla) is prepped <b>last</b> |
| Extent                  | Wide enough to allow <b>extension of the incision, new incisions and drain sites</b> without re-prepping                                                                                         |
| Contraindications       | Chlorhexidine — <b>middle ear (ototoxic), eye (keratitis), neural tissue</b> ; iodophors — neonates, iodine allergy                                                                              |

## 2.3 Immediate pre-transfer preparation

- Patient **voids** immediately before transfer; catheterise only if indicated, and **remove within 24 h** (CAUTI prevention).
- **Remove and document the disposition of:** dentures/bridges, contact lenses, jewellery (tape a wedding ring if it cannot be removed), **nail polish and artificial nails** (interfere with pulse oximetry and harbour organisms), hairpins, prostheses, valuables — exactly what Form 1-1 records.
- **Antibiotic prophylaxis: single IV dose within 60 minutes of skin incision** (120 min for vancomycin); re-dose if surgery exceeds 2 half-lives or blood loss > 1500 mL; **stop within 24 h**. For tourniquet surgery the infusion must finish **before inflation**.

- **Site marked** by the operating surgeon with an indelible marker, with the awake patient participating (§3).
- Confirm **consent, allergies, fasting status, prostheses/implants, blood availability** and that imaging and old records are in the OR.

### 3. ADMISSION PROCEDURE & PATIENT IDENTIFICATION

The PDF teaches the Joint Commission Universal Protocol in full. The **WHO Surgical Safety Checklist** is absent, and it is the internationally examined instrument.

#### 3.1 WHO Surgical Safety Checklist (2008) — 3 phases, 19 items

| Phase              | When                                         | Core content                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. SIGN IN</b>  | Before induction of anaesthesia              | Patient confirms <b>identity, site, procedure, consent</b> • site marked • anaesthesia safety check • <b>pulse oximeter on and working</b> • known allergy • difficult airway / aspiration risk • risk of blood loss > 500 mL (7 mL/kg in children) → access and fluids planned                                                                                       |
| <b>2. TIME OUT</b> | After induction, <b>before skin incision</b> | All members <b>state name and role</b> • surgeon, anaesthetist and nurse verbally confirm patient, site, procedure • <b>anticipated critical events</b> (critical steps, duration, blood loss; anaesthetic concerns; sterility indicator and equipment issues) • <b>antibiotic given in the last 60 min?</b> • imaging displayed • VTE prophylaxis • correct position |
| <b>3. SIGN OUT</b> | Before the patient leaves the OR             | <b>Name of the procedure recorded • instrument, sponge and needle counts correct • specimens labelled and read aloud • equipment problems • key concerns for recovery</b>                                                                                                                                                                                             |

#### DISTINGUISH THESE THREE — favourite MCQ

**Universal Protocol (JC, July 2004)** = pre-procedure **verification** + **site marking** + **‘time out’** immediately before starting. Aim: prevent wrong-site, wrong-procedure, wrong-person surgery.

**AORN (March 2005)** added confirmation of the **correct patient position** to the time out.

**WHO Checklist (2008)** = the **3-phase, 19-item** tool; broader — it also covers airway, blood loss, antibiotics, imaging, counts and specimen labelling. It cut mortality from 1.5 % to 0.8 %.

**Site marking:** by the **operating surgeon, indelible marker**, at/near the incision, **with the awake patient participating, before** entering the OR; mark **laterality and level**; never mark the non-operative side.

**At least two identifiers**, and the patient **states** their own name — never “Are you Mary Jones?”

### 4. TRANSFER PROCEDURE & POSITION

#### 4.1 Safe transfer — the numbers

- Minimum **two persons**; **≥ 4 or a mechanical lifter** for an anaesthetised, obese or unstable patient; **one person on each side** of the table; **lock the wheels** of gurney and table.
- Devices that reduce the load: **roller board (Davis), slide sheet, Hoyer pad, air-assisted transfer mattress**.
- **NIOSH maximum recommended lift = 51 lb (23 kg)**; for **patient handling the limit is 35 lb (16 kg)** — above that, use an aid or a lift team. **Back injury is the commonest occupational injury in health care**.
- **Safety strap across the mid-thighs, ≈ 3 inches (7.5 cm) above the knees** — the single most important measure against a fall; two fingers should slide under it.

- Protect during transfer: **airway/ET tube, IV, arterial and CVP lines, epidural, Foley, drains** — and the **fingers** when table sections are raised.
- **Never move an anaesthetised patient without the anaesthetist's permission and direction.** The anaesthetist guards head/airway/neck, the surgeon protects fractures, the circulator applies padding and the strap.

## 4.2 Position-specific complications — the nerve-injury map

The PDF describes each position well but omits the complications, which is what gets asked.

| Position                                         | Nerve / structure at risk                                                                                                                                      | Prevention                                                                                                                                                                              | Physiological effect                                                                                                                    |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supine</b>                                    | <b>Ulnar nerve — commonest peri-operative nerve injury;</b> brachial plexus; radial n.; occiput, sacrum, heels                                                 | Arms abducted $\leq 90^\circ$ , <b>palms up (supinated)</b> , elbows padded, legs uncrossed                                                                                             | ↓ FRC; aorto-caval compression in late pregnancy                                                                                        |
| <b>Trendelenburg</b>                             | Brachial plexus (shoulder braces), retinal/cerebral congestion, sliding                                                                                        | Braces over the <b>acromio-clavicular area, never on the soft neck</b> ; limit to 30–45° and to the shortest time                                                                       | <b>↑ venous return, ↑ ICP, ↑ IOP, ↑ CVP; ↓ FRC and compliance → atelectasis;</b> facial/laryngeal oedema; tube migration; regurgitation |
| <b>Reverse Trendelenburg / sitting</b>           | Sciatic stretch, foot drop, venous pooling; <b>venous air embolism</b> in the sitting position                                                                 | Padded footboard, antiembolism stockings + compression device, slow position change                                                                                                     | <b>↓ venous return → hypotension; ↓ cerebral perfusion</b>                                                                              |
| <b>Lithotomy</b>                                 | <b>Common peroneal nerve — commonest here → foot drop;</b> femoral, obturator, saphenous, sciatic; <b>lower-limb compartment syndrome</b>                      | Pad the <b>fibular head</b> away from the post; <b>raise and lower both legs together, slowly, by two people</b> ; avoid extreme abduction; <b>limit time (risk rises beyond 2–4 h)</b> | On <b>lowering the legs → sudden ↓ venous return → hypotension; ↓ FRC</b>                                                               |
| <b>Prone</b>                                     | Brachial plexus, ulnar; <b>eye → corneal abrasion and post-operative visual loss (POVL) from ischaemic optic neuropathy</b> ; breasts, genitalia, iliac crests | <b>Chest rolls from clavicle to iliac crest</b> ; head on a padded donut/horseshoe with the <b>eyes free of all pressure</b> ; neutral neck; re-check the ET tube after turning         | ↑ abdominal and epidural venous pressure → bleeding; ↓ cardiac index                                                                    |
| <b>Lateral (kidney / thoracotomy)</b>            | Dependent brachial plexus and axillary vessels, dependent common peroneal and ear                                                                              | <b>Axillary roll caudal to the axilla — never in it</b> ; pillow between the legs; head in line with the spine                                                                          | <b>V/Q mismatch — dependent lung better perfused but less ventilated</b> ; kidney rest ↓ venous return                                  |
| <b>Jackknife / Kraske (a PRONE modification)</b> | Face, eyes, genitalia, knees                                                                                                                                   | Padding at hips, pillow under pelvis and shins                                                                                                                                          | Marked ↓ venous return and ↓ FRC                                                                                                        |
| <b>Fracture table</b>                            | <b>Perineal post → pudendal nerve injury</b> , perineal pressure necrosis                                                                                      | Well-padded post, minimum effective traction, protect the well leg; <b>no skin-metal contact</b> (ESU burn)                                                                             | Blood loss; fat-embolism risk                                                                                                           |

## POSITIONING — GOLDEN RULES

Position is determined by the **surgical approach + patient's condition + anaesthetic technique + surgeon's preference**. Positioning aids must be **in the room before the patient arrives**.

**PADDING** — Pressure points padded, **A**lignment maintained, **D**ignity/exposure minimised, **D**evelopments ready beforehand, **I**V lines protected, **N**erves free of stretch and compression, **G**entle simultaneous movement. Modifications of **supine**: Trendelenburg, reverse Trendelenburg, Fowler's/sitting, lithotomy. Modification of **prone**: Kraske/jackknife.

**Compartment syndrome after lithotomy** = pain out of proportion, tense swollen calf, paraesthesia → **fasciotomy**. Always examine and document limb movement and paraesthesia on arrival in PACU.

## 5. ENVIRONMENTAL CONTROLS

### 5.1 Operating room design — the numbers the PDF omits

| Parameter                 | Standard                                                              | Exam point                                                                                              |
|---------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Temperature               | <b>20-23 °C (68-75 °F)</b>                                            | Team comfort vs <b>patient hypothermia</b> ; raise for neonates and burns                               |
| Relative humidity         | <b>20-60 %</b> (classically 30-60 %)                                  | <b>Too low → static spark risk; too high → microbial growth and condensation</b>                        |
| Air changes per hour      | Minimum <b>15</b> , recommended <b>20-25</b> ( $\geq 3$ outdoor air)  | Dilutes contaminants and waste anaesthetic gas                                                          |
| Air pressure              | <b>POSITIVE</b> relative to corridors                                 | Air flows <b>out</b> of the OR. Contrast: <b>TB isolation = NEGATIVE pressure</b>                       |
| Airflow                   | Unidirectional <b>downward</b> over the table, exhaust at floor level | Keeps a clean air column over the sterile field                                                         |
| <b>HEPA filter</b>        | $\geq 99.97$ % of particles $\geq 0.3 \mu\text{m}$                    | The single most asked number. <b>ULPA = 99.999 % at <math>0.1 \mu\text{m}</math></b> (smoke evacuators) |
| Laminar / ultraclean flow | <b>300-600 air changes/h</b>                                          | Used for <b>joint arthroplasty and implant surgery</b>                                                  |
| Noise                     | < 45 dB recommended                                                   | Silence is mandatory during <b>induction</b> — <b>hearing is the last sense to be lost</b>              |

### 5.2 Zoning of the OT complex

| Zone                      | Also called     | Areas                                                         | Attire                                                                         |
|---------------------------|-----------------|---------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Protective / outer</b> | Unrestricted    | Reception, changing rooms, offices, pre-op holding, PACU      | Street clothes permitted                                                       |
| <b>Clean</b>              | Semi-restricted | Peripheral corridors, storage, utility, instrument processing | <b>Scrub attire + head cover</b> ; mask not required                           |
| <b>Sterile / aseptic</b>  | Restricted      | Operating room, scrub area, sterile store                     | Scrub attire + head cover + <b>mask</b> ; sterile gown and gloves in the field |
| <b>Disposal / dirty</b>   | Dirty corridor  | Dirty utility, sluice, disposal corridor                      | Scrub attire + PPE                                                             |

### 5.3 Cleaning — and why fumigation is obsolete

- **Between cases:** all horizontal surfaces, the table, positioning devices, lights, ESU, suction and the floor **within 1-1.5 m of the field** are wiped with a hospital germicide. The PDF's key point: **every case is cleaned identically** — special 'ritual' cleaning after 'dirty' cases is **obsolete**, because all cases are treated as potentially infectious.
- **Terminal (daily) clean:** whole room, walls, ceiling, equipment wheels, scrub sinks, corridors, storage; wet-vacuum the floor.
- **Formaldehyde fumigation is no longer recommended** — Group 1 carcinogen, mucosal irritant, needs the room sealed and idle 24–48 h, and penetrates crevices unreliably. Modern no-touch options: **vaporised hydrogen peroxide, chlorine dioxide, UV-C** — always **after** physical cleaning, never instead of it.
- **Routine environmental culturing is not recommended** — only for outbreak investigation or after construction.
- **Blood spill:** absorb, then disinfect with **1 % sodium hypochlorite (10 000 ppm)** for 10 min.

### 5.4 Levels of disinfection (needed for §9 as well)

| Level                   | Kills                                                                               | Agents                                                                                                        | Use                                                                           |
|-------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Sterilization</b>    | <b>All microbes including spores</b>                                                | Steam, EO, H <sub>2</sub> O <sub>2</sub> plasma, glutaraldehyde 10 h, peracetic acid, radiation               | <b>Critical</b> items                                                         |
| <b>High-level (HLD)</b> | All except large numbers of spores                                                  | Glutaraldehyde 2 %, <b>OPA 0.55 % (12 min)</b> , H <sub>2</sub> O <sub>2</sub> 7.5 % (30 min), peracetic acid | <b>Semi-critical</b> — endoscopes, laryngoscope blades, respiratory equipment |
| <b>Intermediate</b>     | Vegetative bacteria, <b>M. tuberculosis</b> , most viruses/fungi; <b>not spores</b> | Alcohols 60–90 %, hypochlorite, iodophors, phenolics                                                          | Soiled non-critical surfaces                                                  |
| <b>Low</b>              | Most vegetative bacteria; <b>not TB, not spores</b>                                 | Quaternary ammonium compounds                                                                                 | <b>Non-critical</b> — floors, walls, BP cuffs, furniture                      |

**Key agent facts:** alcohol — fastest kill, no residual action, not sporicidal, inactivated by organic matter, flammable. Chlorhexidine — persistent residual activity, not inactivated by blood, but ototoxic and corneal-toxic. Povidone-iodine — broad but inactivated by blood/pus and needs contact time to release free iodine. Hypochlorite — broad including spores, but corrosive and never mixed with acids or ammonia (chlorine gas). Quaternary ammonium compounds — inactivated by soap, hard water and gauze.

### 5.5 Aseptic technique and the sterile field — hard rules

1. Only **sterile items** are used in a sterile field.
2. The gown is sterile **in front from chest to the level of the field**, and the sleeves from **5 cm above the elbow to the cuff**. The **back is never sterile**.
3. A **draped table is sterile only at table-top level** — anything below the edge is unsterile and is **never brought back up**.
4. Sterile persons keep hands **at or above waist level, in front of the body**, and pass **back-to-back** or **face-to-face**.
5. Unsterile persons stay **≥ 30 cm (1 ft)** away, never reach over the field, and always **face** it.

6. The **edge of any wrapper, package or basin is unsterile** — conventionally a **2.5 cm (1 inch)** margin.
7. **Moisture contaminates by capillary action (strike-through)** — a wet field is a contaminated field.
8. Packages are opened **farthest corner first, nearest corner last**, held away from the body, so you never reach over the contents.
9. Sterile items stay **in view**; anything of **doubtful sterility is unsterile** ('when in doubt, throw it out').

## 5.6 Surgical hand antisepsis

| Item             | Standard                                                                                                                                                                                                                |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agents           | <b>4 % chlorhexidine gluconate</b> or <b>7.5 % povidone-iodine</b> , or an alcohol-based surgical hand rub (WHO now prefers the rub)                                                                                    |
| Duration         | <b>First scrub of the day 3-5 minutes</b> ; older counted-brush-stroke method = 30 strokes to nails, 20 per surface                                                                                                     |
| Sequence         | <b>Nails → fingers and interdigital spaces → hands → wrists → forearms to 5 cm above the elbow</b> ; the <b>cleanest part (hand) is held highest</b> ; rinse fingertips to elbow; <b>never return to a cleaned area</b> |
| Before scrubbing | Remove rings, watches, bracelets; nails short and clean, <b>no polish, gel or artificial nails</b> (AORN — they harbour Gram-negatives and fungi); <b>do not scrub with cuts, abrasions or dermatitis</b>               |
| After scrubbing  | Hands <b>above the elbows, in front of the body, above waist and below shoulder level</b> ; dry with a sterile towel from fingers to elbow, never back                                                                  |
| WHO 5 Moments    | <b>1</b> before touching a patient • <b>2</b> before a clean/aseptic procedure • <b>3</b> after body-fluid exposure risk • <b>4</b> after touching a patient • <b>5</b> after touching patient surroundings             |
| Rub vs soap      | Alcohol rub routinely; <b>soap and water is mandatory for visibly soiled hands and for Clostridioides difficile</b> (alcohol is not sporicidal)                                                                         |
| Double gloving   | Reduces inner-glove perforation by $\approx$ <b>70-80 %</b> ; change every 90-150 min and between patients                                                                                                              |

## 5.7 Surgical site infection — classification

| Wound class                    | Definition                                                                                                   | Example                                                          | SSI risk        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------|
| <b>I - Clean</b>               | No inflammation; respiratory/GI/GU tract <b>not entered</b> ; closed primarily                               | Thyroidectomy, hernia repair, joint replacement                  | <b>&lt; 2 %</b> |
| <b>II - Clean-contaminated</b> | Tract entered under <b>controlled conditions</b> , no unusual contamination                                  | Elective cholecystectomy, elective bowel resection, hysterectomy | <b>3-10 %</b>   |
| <b>III - Contaminated</b>      | Fresh accidental wound; <b>major break in technique</b> ; gross GI spillage; acute non-purulent inflammation | Fresh trauma, enterotomy with spillage                           | <b>15-20 %</b>  |
| <b>IV - Dirty / infected</b>   | Old wound with devitalised tissue; existing infection or <b>perforated viscus</b>                            | Faecal peritonitis, perforated appendix                          | <b>30-40 %</b>  |

- **CDC depth classification: superficial incisional** (skin/subcutaneous, within **30 days**); **deep incisional** (fascia/muscle, within **30 days, or 90 days if an implant** is present); **organ/space**.
- Commonest organism = **Staphylococcus aureus**; commonest **source of contamination = the patient's own endogenous flora**.
- **Proven SSI-prevention bundle**: CHG-alcohol prep, no razor shaving, antibiotic within 60 min, **normothermia > 36 °C, glucose < 180 mg/dL**, adequate tissue oxygenation, minimal traffic, good haemostasis and no dead space.

## 6. ELECTRO SURGERY

The PDF gives 5 precautions. The mechanism and the injury mechanisms are what carry the marks.

### 6.1 Fundamentals

| Concept                           | Detail                                                                                                                                                                                                                                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Principle                         | <b>High-frequency alternating current</b> concentrated at a small active electrode; tissue resistance converts it to heat → <b>cutting, coagulation or desiccation</b> . It is <b>not cautery</b> (true cautery = a passively heated wire, no current through the patient) |
| Frequency                         | <b>300 kHz - 3 MHz (radiofrequency)</b> . Above ~ <b>100 kHz</b> the current <b>does not stimulate nerve or muscle (the Faradic effect is avoided)</b> — which is why RF current cuts but does not electrocute                                                             |
| Cut waveform                      | <b>Continuous, low-voltage, high-current</b> (100 % duty cycle) → vaporisation, little haemostasis                                                                                                                                                                         |
| Coag waveform                     | <b>Interrupted, high-voltage, low duty cycle (~6 %)</b> → coagulum. High voltage → <b>greater risk of capacitive coupling, insulation breakdown and sparking</b>                                                                                                           |
| Monopolar circuit                 | Generator → <b>active electrode (pencil)</b> → patient → <b>dispersive/return pad</b> → generator. Current crosses the whole body; <b>the pad is essential</b>                                                                                                             |
| Bipolar circuit                   | Both electrodes are the <b>two tines of the forceps</b> ; current passes only between them. <b>No pad needed</b> , minimal lateral spread, usable in fluid, <b>safest with pacemakers</b> — neurosurgery, ophthalmic, microsurgery                                         |
| Pad placement                     | <b>Clean, dry, hair-free, well-vascularised muscle mass, as close as practicable to the site</b> , whole surface in uniform contact, <b>away from ECG electrodes</b> , bony prominences, scar, tattoos, metal implants and pooled fluid                                    |
| REM / contact-quality monitoring  | A <b>split pad</b> whose two halves are monitored for impedance; the generator <b>shuts down if contact becomes inadequate</b> — has virtually eliminated pad-site burns. Modern generators are <b>isolated</b> , not ground-referenced                                    |
| AEM (active-electrode monitoring) | Shielded laparoscopic instruments that drain stray current — prevents insulation-failure and capacitive-coupling burns                                                                                                                                                     |

### 6.2 The five mechanisms of electrosurgical injury

- 1. Return-electrode (pad-site) burn** — poor or partial pad contact, hair, dried gel, pad over bone or scar. Prevented by correct placement + REM.
- 2. Alternate-site burn** — the patient's skin touches a **metal surface** (table edge, stirrup, armboard screw) which becomes an unintended return path. The PDF states this rule twice: **no skin-to-metal contact**.
- 3. Direct coupling** — the activated electrode touches another metal instrument (laparoscope, retractor), which burns adjacent tissue.
- 4. Capacitive coupling** — current crosses **intact insulation** by an electrostatic field; classic setting = a **metal instrument through a plastic (hybrid) trocar**, or coiled cords. Worse with **coag mode and long thin instruments**.
- 5. Insulation failure** — a pin-hole break delivers current **outside the surgeon's view** → **delayed bowel perforation 3-10 days post-operatively**. Inspect insulation before every use.



### 6.3 Other examination points

- **Repeated requests to increase the power** = poor pad contact, a loose connection, a frayed cord or a failing generator. **Stop, check the circuit and pad; do not simply turn it up.** Replace and label the unit and **document the serial numbers and the reason.**
- **Keep the pencil in its holster** when not in use; clean eschar with a non-abrasive pad.
- **Pacemaker / ICD:** prefer **bipolar**; if monopolar is essential use short bursts at low power with the current path away from the device, keep the **crash cart with defibrillator immediately available and document that it is there**, monitor ECG and pulse continuously, and **interrogate the device afterwards.**
- **Never use ESU or laser in an oxygen-enriched field** (mouth, oropharynx, tracheostomy, under drapes with high-flow O<sub>2</sub>) — pause the oxygen and allow washout.
- **Advanced devices in one line each:** **ultrasonic (Harmonic) scalpel** — mechanical vibration at **55.5 kHz**, cavitation at a **lower temperature (50-100 °C)**, no current through the patient; **bipolar vessel sealing (LigaSure)** — seals vessels up to **7 mm**; **argon-enhanced** coagulation — thin uniform eschar, gas-embolism risk.
- **Surgical smoke (plume):** > 150 chemicals plus viable cells and viral DNA (**documented HPV transmission to surgeons**). **Ablating 1 g of tissue ≈ 3-6 unfiltered cigarettes.** Control = **smoke evacuator with an ULPA filter (99.999 % at 0.1 µm) held within 2.5-5 cm (1-2 in) plus a high-filtration mask or N95** — a standard mask is **not** adequate.

#### EQUIPMENT NUMBERS THAT SHARE THIS SUB-TOPIC

**Pneumatic tourniquet:** cuff ends overlap **3-6 inches**; **never at or near the elbow**; **upper limb 250-300 mmHg for ≤ 60 min**, **lower limb 300-350 mmHg for ≤ 90-120 min**; **deflate 10-15 min** if longer; exsanguinate with an Esmarch bandage **before** inflation; **record site, pressure and both inflation and deflation times.** On release: ↓ BP, transient acidosis, ↑ K<sup>+</sup>, ↑ EtCO<sub>2</sub>.

**Laser:** **wavelength-specific eye protection for everyone including the patient**; warning signs on doors, windows covered, key held by the laser safety officer, **standby mode when not firing**, **foot pedal only to the surgeon**, **matte instruments**, **wet drapes**, **saline available**, **plume evacuator**, **laser-safe ET tube with a saline-filled cuff**, and **never in an oxygen-enriched field.**

**Powered saws/drills:** engineering clearance, test before use, attachments secure and sharp, **let the rotating part come to a complete standstill before setting it down**, face shield mandatory.

## 7. OPERATIVE RECORDS

### 7.1 Specimen handling — a favourite MCQ the PDF omits

| Specimen               | Handling                                                                                                                                                              |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Routine histopathology | <b>10 % neutral buffered formalin</b> , volume ≈ <b>10× the specimen</b> ; container labelled with name, ID, date, site/side; <b>label read aloud during Sign Out</b> |
| <b>Frozen section</b>  | <b>FRESH</b> , in <b>saline-moistened gauze</b> — <b>NEVER in formalin</b> (formalin prevents further study and immunohistochemistry); send immediately               |
| Culture / microbiology | <b>Sterile container, no fixative</b> ; anaerobic transport for anaerobes; <b>never formalin</b>                                                                      |
| Stones/calculi         | Dry container, <b>no fixative</b>                                                                                                                                     |
| Implants               | Record <b>type, manufacturer, size, serial/lot number</b> in the intra-operative record and the <b>implant log</b> (traceability for recalls)                         |
| Lost specimen          | A <b>serious, irreplaceable incident</b> — incident report mandatory                                                                                                  |

## 7.2 What makes the record legally sound

- **JC-mandated minimum content:** surgeon's name and assistants, **pre-operative diagnosis**, procedure performed, **post-operative diagnosis**, findings, **specimens removed (type and number)**, estimated blood loss, and the names of all team members giving intra-operative care.
- Also document: the **time out**, the **counts and their outcome**, **skin condition before and after prep** and the solution used, position and positioning aids, **ESU pad site, serial number and settings**, tourniquet data, drains/catheters/packing, medications and irrigations (name, strength, dose, route, time, who gave them), implants, intake/output, transfusion checks, and any **unusual event with the actions taken**.
- **Rules:** contemporaneous, legible, factual and objective (no opinion or blame), signed and timed; **correct by a single line-through with initials, date and time — never erase, overwrite or use correction fluid**; leave no blank spaces; label late entries.
- **Incident (occurrence) report** for: incorrect count, medication error, burn, fall, retained item, equipment failure, needle-stick, lost specimen, unplanned re-operation and **death in the OR**.
- **Only what is documented is a legal fact** — the most quoted line in your PDF (pp. 8 and 48). An undocumented event 'did not happen' in court.

## 7.3 Consent — the parts the forms do not explain

| Point                                     | Rule                                                                                                                                                                                                            |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Five elements of valid consent            | <b>Disclosure • Capacity • Voluntariness • Comprehension • Documentation</b>                                                                                                                                    |
| Must be disclosed                         | Diagnosis, nature and purpose of the procedure, risks and benefits, <b>alternatives including doing nothing</b> , and consequences of refusal. <b>No guarantee of outcome may be given</b>                      |
| Timing                                    | <b>Before sedative premedication</b> — hence the wording on Form 1-2, “I am not under the influence of any preoperative or other sedating medication”                                                           |
| Age                                       | <b>18 years</b> in India ( <b>IPC §§ 87-88</b> ). For a child <b>under 12</b> or a person of unsound mind, the guardian consents ( <b>IPC § 89</b> )                                                            |
| Emergency / unconscious                   | <b>Doctrine of necessity — IPC § 92:</b> life-saving treatment in good faith without consent when the patient cannot consent and no representative is available                                                 |
| Consent obtained by fear or misconception | <b>Invalid — IPC § 90</b>                                                                                                                                                                                       |
| Special consent needed for                | <b>Sterilization</b> (patient must state understanding of permanent, <b>irreversible</b> loss of fertility — Form 1-3), blood products, photography, implants, organ donation, HIV testing                      |
| Who obtains vs who verifies               | The <b>operating surgeon obtains</b> it. The perioperative practitioner <b>verifies</b> that a valid, signed, dated consent exists and matches the procedure, site and side, and <b>reports any discrepancy</b> |
| Withdrawal                                | May be withdrawn at <b>any time</b> while the patient is competent                                                                                                                                              |
| Blood refusal                             | Honour it, <b>document it on the checklist</b> , and plan cell-saver/tranexamic acid                                                                                                                            |

## MEDICO-LEGAL DOCTRINES — one-liners

**Negligence — the 4 D's:** Duty, Dereliction (breach of the standard of a reasonably competent practitioner — the **Bolam test**), Direct causation, Damage.

**Res ipsa loquitur** ('the thing speaks for itself') — negligence presumed; classic examples are a **retained sponge or instrument**, wrong-site surgery, and a diathermy burn.

**Vicarious liability / respondeat superior** — the hospital is liable for its employees' acts.

**Battery** = operating with **no** consent. **Negligence** = operating with **inadequately informed** consent.

## 8. COUNTING PROCEDURE

Learn the 9 counting rules from the PDF. What it lacks is the terminology and epidemiology.

- A retained item is a **'never event' / sentinel event**. The **commonest retained item is a sponge/gauze (≈ 50-70 %)**, then instruments and needles; commonest site = **abdomen/pelvis**.
- A retained gauze encased in a foreign-body granuloma = **gossypiboma (textiloma)** — presents weeks to years later with a mass, pain, sinus, abscess, obstruction or fistula, and is a classic **res ipsa loquitur** claim.
- **Risk factors:** emergency surgery, **unexpected intra-operative change of procedure**, high BMI, multiple teams or staff changes, and omitted or incorrect counts.
- **Counts are taken:** (1) **before the procedure (baseline)**, (2) when **items are added**, (3) **before closure of a cavity within a cavity** (uterus, bladder, bowel), (4) when **wound/fascial closure begins**, (5) at **skin closure**, and (6) at **any change of scrub or circulating personnel**.
- **The circulator records the count — that record is a legal responsibility.** Both persons count **aloud together** as the scrub person touches each item.
- All sponges used in a wound must be **radio-opaque (X-ray detectable)**; **X-ray-detectable sponges are never used as dressings**, and dressing sponges are never used in the wound. Adjuncts: **bar-coded counting and radiofrequency (RF) tag detection wands**.
- **If the count is incorrect:** repeat it → **inform the surgeon**, who explores the wound → if still missing, **intra-operative X-ray before the patient leaves the OR** → notify the OR supervisor → **document and file an incident report**. **Nothing (linen, trash, instruments, sponges) leaves the room until the final count is correct.**

## 9. STERILIZATION — HIGHEST-YIELD SECTION IN THE UNIT

### 9.1 Definitions — get these exactly right

| Term                   | Definition                                                                                                                    |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Sterilization</b>   | Destruction or removal of <b>all forms of microbial life, including bacterial and fungal spores</b> . An <b>absolute</b> term |
| <b>Disinfection</b>    | Destruction of <b>pathogens but not necessarily spores</b> ; used on <b>inanimate</b> objects                                 |
| <b>Antisepsis</b>      | Destruction/inhibition of organisms on <b>living tissue</b>                                                                   |
| <b>Decontamination</b> | Making an item safe to handle — <b>always the FIRST step</b> , before disinfection or sterilization                           |
| <b>Cleaning</b>        | Physical removal of soil — <b>a dirty instrument cannot be sterilized</b> (organic matter blocks the sterilant)               |

| Term                                   | Definition                                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Asepsis</b>                         | <b>Medical asepsis</b> = clean technique (reduce numbers); <b>surgical asepsis</b> = sterile technique (absence of all organisms) |
| <b>Sporicidal</b>                      | Kills spores — the property that makes an agent a <b>sterilant</b> rather than a disinfectant                                     |
| <b>SAL (Sterility Assurance Level)</b> | <b>10<sup>-6</sup></b> — probability of $\leq$ one viable organism in 10 <sup>6</sup> items. The definition of ‘sterile’          |
| <b>D-value</b>                         | Time or dose that reduces the population by <b>1 log<sub>10</sub> (90 %)</b>                                                      |

## 9.2 Spaulding classification

| Category             | Definition                                                | Examples                                                                              | Processing                                 |
|----------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Critical</b>      | Enter <b>sterile tissue or the vascular system</b>        | Surgical instruments, implants, needles, cardiac and urinary catheters, laparoscopes  | <b>Sterilization</b>                       |
| <b>Semi-critical</b> | Contact <b>intact mucous membranes or non-intact skin</b> | Flexible endoscopes, laryngoscope blades, ET tubes, respiratory/anaesthesia equipment | <b>High-level disinfection</b>             |
| <b>Non-critical</b>  | Contact <b>intact skin only</b>                           | BP cuffs, stethoscopes, OR table, floors, furniture                                   | <b>Low/intermediate-level disinfection</b> |

Resistance to killing, most → least: **Prions > bacterial spores > coccidia > mycobacteria > small non-lipid viruses > fungi > vegetative bacteria > lipid-enveloped viruses (HIV, HBV — the EASIEST to kill)**. A common MCQ inverts the last point. **Prions: resist routine autoclaving — need 1 N NaOH plus 134 °C for 18 min, and single-use instruments where possible.**

## 9.3 Moist heat — saturated steam under pressure (autoclave)

**Mechanism: denaturation and coagulation of proteins. Moist heat beats dry heat because latent heat of vaporisation is released when steam condenses. It is the method of choice — fastest, cheapest, most reliable, non-toxic.**

| Cycle                          | Temperature            | Pressure           | Hold time                                           | Use                                                                                 |
|--------------------------------|------------------------|--------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Gravity displacement</b>    | <b>121 °C (250 °F)</b> | <b>15 psi</b>      | <b>15-20 min</b> (wrapped 30 min)                   | General load, glassware, rubber, media                                              |
| Gravity — high temp            | <b>134 °C</b>          | <b>30 psi</b>      | <b>3-3.5 min</b>                                    | Rapid cycles                                                                        |
| <b>Prevacuum (high-vacuum)</b> | <b>132-135 °C</b>      | <b>≈ 30 psi</b>    | <b>3-4 min + drying</b>                             | <b>Wrapped trays, textiles, lumened items</b> — air is actively evacuated           |
| <b>Flash / IUSS</b>            | <b>132 °C (270 °F)</b> | <b>≈ 27-30 psi</b> | <b>3 min unwrapped metal; 10 min porous/lumened</b> | <b>Only an urgently needed unwrapped single item</b> — never implants or whole sets |
| Prions (CJD)                   | 134 °C                 | —                  | <b>18 min</b>                                       | After NaOH immersion                                                                |

- **Air is the enemy of the autoclave** — trapped air lowers the temperature at a given pressure. So: **do not overload, do not wrap tightly, use mesh-bottom trays, place packs on edge with space between them, and never position a bowl so that it traps air.**

- **Bowie-Dick test** = a **daily test of air removal / steam penetration in a PREVACUUM sterilizer**, run in an **empty chamber at 134 °C for 3.5 min**. It is a **Class 2 chemical indicator** and it does **not** prove sterility.
- **Biological indicator at least weekly (daily preferred) and with EVERY implant load** — implant loads are quarantined until the result is available. Vacuum leak test weekly.
- **Cannot be autoclaved:** sharp cutting edges (blunted), oils, greases, powders, anhydrous materials, low-melting plastics, lensed endoscopes, heat-labile electronics.
- Drying is part of the cycle — **a wet pack is a contaminated pack**; do not touch or move packs until cool and dry.

## 9.4 Dry heat (hot air oven)

**Mechanism: oxidation — needs much higher temperature and longer time than moist heat.**

| Temperature   | Holding time                               |
|---------------|--------------------------------------------|
| <b>160 °C</b> | <b>2 hours</b> (the pair most often asked) |
| <b>170 °C</b> | <b>1 hour</b>                              |
| <b>180 °C</b> | <b>30 minutes</b>                          |

- **Uses:** glassware, **sharp instruments (does not blunt or corrode them)**, oils, greases, powders, anhydrous fats, glass syringes. **Not for** rubber, plastics or fabrics.
- Load loosely for air circulation; **allow to cool before opening**.
- **Other dry-heat methods:** **incineration** ( $\geq 1100$ – $1200$  °C, final disposal of contaminated and anatomical waste), **red-heat flaming** (loops), **glass-bead steriliser** (250 °C, 20–40 s).
- **Biological indicator: Bacillus atrophaeus.**

## 9.5 Ethylene oxide gas

| Parameter                       | Value                                                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanism                       | <b>Alkylation</b> of DNA, RNA and protein                                                                                                                                          |
| <b>The 4 critical variables</b> | <b>Gas concentration 450–1200 mg/L • temperature 37–63 °C • relative humidity 40–80 % • exposure 1–6 h</b> (the PDF quotes 3–7 h cycles) — <b>all four must be met</b>             |
| <b>Aeration is mandatory</b>    | <b>Mechanical aerator 8–12 h at 50–60 °C</b> (air changed $\geq 4$ times/h), <b>or a ventilated room for 7 days at 18–22 °C</b> . Residual EO causes chemical burns and haemolysis |
| Uses                            | <b>Heat- and moisture-sensitive items</b> — flexible endoscopes, cardiac catheters, plastics, electrical and electronic equipment, prosthetic valves, pacemakers, delicate optics  |
| Rule                            | <b>Anything that can withstand steam should NOT be gas-sterilized</b>                                                                                                              |
| Hazards                         | <b>Toxic, irritant, mutagenic, teratogenic, carcinogenic (leukaemia/lymphoma), flammable and explosive; OSHA PEL 1 ppm (8-h TWA)</b> , excursion 5 ppm/15 min                      |
| Biological indicator            | <b>Bacillus atrophaeus</b>                                                                                                                                                         |
| Note on the indicator           | An EO chemical indicator shows the item was <b>exposed</b> to EO — it does <b>not</b> prove sterilization occurred                                                                 |

## 9.6 Low-temperature hydrogen peroxide gas plasma (STERRAD)

- **58 % hydrogen peroxide** vaporised under vacuum, then excited by **radiofrequency energy** into a plasma of free radicals → kills bacteria, fungi, viruses **and spores**.

- **45-50 °C, cycle ≈ 1 hour; by-products are only water vapour and oxygen → NO aeration or cooling needed**, no toxic residue.
- Ideal for **fibre-optic cables, endoscopes, microsurgical and electrical equipment, glass, ceramics, metals**.
- **Cannot be used for cellulose — paper, cotton, linen, gauze, dressings** (they absorb the sterilant) — nor for liquids, powders or long narrow lumens. High capital cost.
- **Biological indicator: Geobacillus stearothermophilus.**

## 9.7 Liquid chemical sterilization / high-level disinfection

| Agent                                         | Sterilization             | High-level disinfection                                                                             | Notes                                                                                                                                                                                                                                            |
|-----------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Glutaraldehyde 2 % (activated — Cidex)</b> | <b>10 hours</b>           | <b>20-45 min at 20-25 °C</b> (your PDF says 10 min — quote the textbook if the question is from it) | <b>Does not corrode lenses or metal</b> → endoscopes. <b>Rinse in sterile distilled water. Test potency with a strip before each use.</b> Irritant/sensitiser — <b>asthma, dermatitis; ceiling 0.05 ppm</b> ; wear polyethylene or double gloves |
| <b>OPA 0.55 %</b>                             | —                         | <b>12 min at 20 °C</b>                                                                              | Faster, less irritant; stains skin grey                                                                                                                                                                                                          |
| <b>Peracetic acid 0.2 % (Steris)</b>          | <b>12 min at 50-56 °C</b> | —                                                                                                   | Single-use, automatic dilution, effective <b>even with organic soil</b> , low surface tension → reaches lumens; <b>decomposes to acetic acid + water + oxygen</b>                                                                                |
| <b>Hydrogen peroxide 7.5 %</b>                | 6 hours                   | <b>30 min</b>                                                                                       | Non-toxic breakdown products                                                                                                                                                                                                                     |

## 9.8 Radiation and filtration

| Method                            | Detail                                                                                                                                                                                                                                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ionising (γ) radiation</b>     | <b>Cobalt-60, dose 25 kGy (2.5 Mrad).</b> A <b>cold, industrial</b> method for pre-packed <b>single-use disposables</b> — syringes, needles, gloves, catheters, sutures, blades, heart valves, bone grafts. <b>Not used inside hospitals.</b> BI = <b>Bacillus pumilus</b> |
| <b>Non-ionising UV-C (254 nm)</b> | <b>Surface and air disinfection only — poor penetration</b> (blocked by glass, paper, water film). A <b>disinfectant, not a steriliser.</b> Hazard: <b>keratoconjunctivitis and erythema</b> — never run with the room occupied                                            |
| <b>Filtration</b>                 | <b>0.22 μm</b> membrane for <b>heat-labile liquids</b> (sera, antibiotic and vaccine solutions, media). <b>Removes but does not kill</b> , and <b>does not retain viruses or mycoplasma</b> (needs 0.1 μm). Air equivalent = <b>HEPA</b> (\$5.1)                           |

## 9.9 Monitoring — the three-indicator system

| Type                         | What it is                                  | What it proves                                                                     | Examples            |
|------------------------------|---------------------------------------------|------------------------------------------------------------------------------------|---------------------|
| <b>Physical / mechanical</b> | Gauges, printout, chart recorder, cycle log | That the <b>machine</b> reached the parameters; read and signed <b>every cycle</b> | Sterilizer printout |

| Type                                     | What it is                                                                                  | What it proves                                                           | Examples                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chemical (ISO 11140, classes 1-6)</b> | Heat/chemical-sensitive dyes on tape, strips, cards                                         | <b>Exposure to the process — NOT sterility</b>                           | <b>Class 1</b> process indicator (external autoclave tape) • <b>Class 2 = Bowie-Dick</b> • <b>Class 3</b> single-variable • <b>Class 4</b> multi-variable • <b>Class 5 integrating</b> (all critical variables, correlates with the BI) • <b>Class 6 emulating</b> (most stringent). An <b>internal</b> indicator goes in the <b>centre of every pack</b> |
| <b>Biological (BI) — GOLD STANDARD</b>   | Calibrated <b>highly resistant bacterial spores</b> processed with the load, then incubated | <b>The only indicator that proves the process actually killed spores</b> | <b>Steam &amp; H<sub>2</sub>O<sub>2</sub> plasma → Geobacillus stearothermophilus</b> (incubate 55–60 °C) • <b>Dry heat &amp; EO → Bacillus atrophaeus</b> (35–37 °C) • <b>γ-radiation → Bacillus pumilus</b>                                                                                                                                             |

### PACKAGING, STORAGE AND THE PROCESSING SEQUENCE

Packaging must **allow sterilant penetration, be a microbial barrier, permit aseptic opening and resist moisture**: double-thickness muslin, non-woven polypropylene, crepe paper, paper-plastic peel pouches, rigid containers with filters. **Never closed metal containers or foil for steam.**

**Shelf-life is EVENT-RELATED, not TIME-RELATED** — a pack is sterile **until an event compromises it** (wet, torn, punctured, soiled, seal broken, dropped, over-handled). The old “sterile for 30 days” rule is obsolete.

Storage: clean, dry, dust-free, closed cupboard, **20–23 °C, humidity 30–60 %, ≥ 20–25 cm above the floor, 45 cm below the ceiling/sprinkler, 5 cm from outside walls**; stock rotated **FIFO**.

**Processing sequence**: point-of-use enzymatic spray in the OR → closed transport → **decontamination/washer-disinfector → ultrasonic cleaner (cavitation reaches box locks, serrations and lumens)** → rinse and dry → inspect, lubricate, assemble with **hinges and box locks OPEN** → pack with an internal indicator → sterilize → cool/dry → store.

## 10. EMERGENCIES AND DISASTERS

### 10.1 Cardiac arrest in the OR — the numbers

| Item                      | Value                                                                                                                                                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sequence                  | <b>C-A-B</b> ; high-quality compressions are the priority                                                                                                                                                                                                       |
| Rate / depth              | <b>100–120 per minute; ≥ 5 cm and ≤ 6 cm</b> in adults (⅓ of chest depth in children)                                                                                                                                                                           |
| Compression : ventilation | <b>30 : 2</b> unintubated adult; <b>continuous compressions with 1 breath every 6 s</b> once an advanced airway is in; <b>15 : 2</b> two-rescuer paediatric                                                                                                     |
| Quality                   | Interruptions < <b>10 s</b> , <b>full recoil</b> , change compressor <b>every 2 min</b> , compression fraction > <b>60–80 %</b> ; <b>EtCO<sub>2</sub> &lt; 10 mmHg = poor CPR, a sudden rise suggests ROSC</b>                                                  |
| Defibrillation            | <b>Biphasic 120–200 J</b> (monophasic 360 J). <b>Shockable = VF and pulseless VT only; PEA and asystole are NOT shockable</b>                                                                                                                                   |
| Drugs                     | <b>Adrenaline 1 mg IV every 3–5 min</b> ; <b>amiodarone 300 mg</b> then 150 mg for refractory VF/pVT                                                                                                                                                            |
| Reversible causes         | <b>4 H's</b> — Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia-metabolic, Hypothermia; <b>4 T's</b> — Tension pneumothorax, Tamponade, Toxins, Thrombosis                                                                                                             |
| Crash cart                | <b>Defibrillator + monitor</b> , airway drawer (laryngoscope, ET tubes, LMA, oral airways, suction, bag-valve-mask), <b>in-date emergency drugs</b> , IV access and fluids, <b>checked and re-sealed after every use and each shift with a signed checklist</b> |
| Paediatric                | <b>PALS</b> competency required; shock <b>2 J/kg then 4 J/kg</b> ; adrenaline <b>0.01 mg/kg</b>                                                                                                                                                                 |
| Documentation             | Every drug, dose, route, time, intervention and the person giving it — <b>undocumented care is legally non-existent</b>                                                                                                                                         |

## 10.2 Malignant hyperthermia — full protocol

| Item          | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Genetics      | <b>Autosomal dominant; RYR1 (ryanodine receptor) gene on chromosome 19</b> ; uncontrolled $\text{Ca}^{2+}$ release → sustained contracture and a <b>hypermetabolic crisis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Triggers      | <b>All volatile inhalational agents + SUCCINYLCHOLINE. Safe: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants, all local anaesthetics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Earliest sign | <b>Unexplained rise in <math>\text{EtCO}_2</math></b> with tachycardia/tachypnoea; <b>masseter spasm</b> after succinylcholine. <b>Hyperthermia is a LATE sign</b> — do not wait for it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Full picture  | Rigidity, hypercarbia, mixed acidosis, <b>hyperkalaemia</b> , arrhythmias, temperature rising 1–2 °C every 5 min, <b>myoglobinuria/rhabdomyolysis</b> , raised CK, DIC, renal failure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Management    | <b>1</b> Call for help and the MH cart • <b>2 stop all triggers</b> , change circuit, <b>hyperventilate with 100 % <math>\text{O}_2</math> at <math>\geq 10</math> L/min</b> • <b>3 DANTROLENE 2.5 mg/kg IV bolus, repeat every 5–10 min up to 10 mg/kg (may need 30)</b> • <b>4 finish/abandon surgery quickly</b> • <b>5 active cooling</b> — iced IV saline, ice packs, cold lavage of cavities; <b>stop at 38 °C</b> • <b>6</b> treat hyperkalaemia and arrhythmias ( <b>avoid calcium-channel blockers with dantrolene</b> ) • <b>7</b> maintain urine output <b>&gt; 1–2 mL/kg/h</b> • <b>8</b> monitor core temperature, ABG, $\text{K}^+$ , CK, coagulation • <b>9 ICU <math>\geq 24</math>–36 h</b> (recrudescence up to 25 %) • <b>10</b> counsel family, refer for <b>caffeine-halothane contracture or genetic testing</b> , issue an alert card |
| Dantrolene    | <b>20 mg lyophilised vials reconstituted with 60 mL sterile water</b> ; keep a ~36-vial stock                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Mortality     | <b>&gt; 70–80 % untreated → &lt; 5–10 % with prompt dantrolene</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## 10.3 Anaphylaxis in the OR

- Commonest triggers: **neuromuscular blocking agents, antibiotics, chlorhexidine, LATEX, colloids, contrast.**
- **Latex anaphylaxis is typically DELAYED 20–60 min** after exposure (absorption across mucosa/peritoneum) — a favourite MCQ. Highest risk: **spina bifida, children with multiple operations, health-care workers. Cross-reacting foods: banana, avocado, chestnut, kiwi.**
- Signs under anaesthesia: **unexplained hypotension and tachycardia (commonest), high airway pressure / bronchospasm, desaturation, flushing, angio-oedema, arrest.**
- **Management: stop the trigger → 100 %  $\text{O}_2$  and secure the airway → ADRENALINE 0.5 mg IM (0.5 mL of 1:1000) into the antero-lateral thigh, repeat every 5 min (or 50  $\mu\text{g}$  IV boluses if monitored and experienced) → rapid IV crystalloid 20 mL/kg, legs up → chlorpheniramine 10 mg, hydrocortisone 200 mg, nebulised salbutamol.** Take **mast-cell tryptase** samples and refer for allergy testing.
- **Latex-safe environment:** first case of the day, latex-free gloves/tubing/catheters, a dedicated **latex-free crash cart**, signage on the door.

## 10.4 Fire in the operating room

| Element                   | Detail                                                                                                             |
|---------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Fire triangle</b>      | <b>FUEL + OXIDISER + IGNITION (heat)</b> — all three abound in an OR                                               |
| Commonest ignition source | <b>Electrosurgical unit (<math>\approx 70</math>–90 % of OR fires)</b> , then laser, then fibre-optic light cables |
| Commonest fuel            | <b>Alcohol-based skin prep</b> (pooled or not dry), then drapes, gowns, gauze, hair, ET tubes                      |



| Element                       | Detail                                                                                                                                                                                                                                                                                                 |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oxidiser</b>               | <b>Oxygen and nitrous oxide</b> — O <sub>2</sub> pools under drapes near the head (the classic head-and-neck fire)                                                                                                                                                                                     |
| Prevention                    | Let alcohol <b>dry ≥ 3 min</b> , never allow pooling, blot puddles; keep FiO <sub>2</sub> < <b>30 %</b> for head/neck work; <b>pause oxygen before activating ESU/laser in the airway</b> ; moisten sponges and drapes; holster the pencil                                                             |
| <b>Response — R A C E</b>     | <b>Rescue</b> • <b>Alarm</b> (call the code, e.g. <b>CODE RED</b> ) • <b>Confine</b> (close doors, shut off medical gases) • <b>Extinguish/Evacuate</b>                                                                                                                                                |
| <b>Extinguisher — P A S S</b> | <b>Pull</b> the pin • <b>Aim</b> at the <b>base</b> of the fire • <b>Squeeze</b> • <b>Sweep</b>                                                                                                                                                                                                        |
| Extinguisher classes          | <b>A</b> ordinary combustibles • <b>B</b> flammable liquids • <b>C electrical</b> → <b>CO<sub>2</sub> or dry chemical, NEVER water</b> • <b>D</b> metals • <b>K</b> kitchen fats. The OR stocks <b>CO<sub>2</sub> or ABC dry chemical</b> ; CO <sub>2</sub> is preferred around equipment (no residue) |
| <b>Airway fire drill</b>      | <b>Stop ventilation and disconnect the oxygen → REMOVE the ET tube and burning material → pour saline into the airway → ventilate on air → re-intubate and bronchoscope → steroids/ICU</b> ; then incident report and device reporting                                                                 |
| Drills                        | <b>At least twice a year, unannounced.</b> Everyone must know the code name and the location of the alarm, extinguishers, gas shut-off valves and evacuation route. <b>Drapes and gowns are fire-RETARDANT, not fire-PROOF</b>                                                                         |

## 10.5 Disaster preparedness and triage

| Concept                            | Content                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Types                              | <b>Internal</b> — fire, power/gas failure, structural, violent intruder; <b>External</b> — earthquake, flood, road/rail crash, blast, chemical leak, epidemic                                                                                                                                                                                                               |
| Framework                          | A written <b>Hospital Disaster Management Plan</b> based on a <b>Hazard Vulnerability Analysis</b> , run through a <b>Hospital Incident Command System</b> , with call-tree, surge capacity and <b>mock drills at least annually</b>                                                                                                                                        |
| <b>START triage</b>                | <b>Simple Triage And Rapid Treatment</b> — ≤ <b>60 s per casualty</b> , assessing <b>RPM: Respiration, Perfusion, Mental status</b>                                                                                                                                                                                                                                         |
| <b>Triage colours</b>              | <b>RED = Priority I, immediate</b> (RR > 30, absent radial pulse / capillary refill > 2 s, or not obeying commands — airway obstruction, major haemorrhage, shock) • <b>YELLOW = Priority II, delayed</b> (can wait 1–4 h) • <b>GREEN = Priority III, minor / walking wounded</b> • <b>BLACK = Priority 0, dead or expectant</b> (not breathing after the airway is opened) |
| Key principle                      | In a mass casualty the aim shifts from ‘the best for each individual’ to <b>‘the greatest good for the greatest number’</b> ; <b>triage is dynamic — re-triage at every stage</b>                                                                                                                                                                                           |
| Power failure                      | Automatic <b>emergency generator</b> back-up, <b>UPS</b> for monitors and ventilators; everyone must know where the <b>high-intensity torches</b> are kept, and where the <b>self-inflating bag and reserve oxygen cylinder</b> are for medical-gas failure                                                                                                                 |
| Emergency signal                   | Each OR has a <b>special emergency signal (usually a flashing red light)</b> operable from within that room                                                                                                                                                                                                                                                                 |
| Code colours (intitution-specific) | Typically <b>Blue</b> cardiac arrest • <b>Red</b> fire • <b>Pink</b> infant abduction • <b>Orange</b> mass casualty • <b>Black</b> bomb threat • <b>Brown</b> evacuation • <b>Grey</b> violent person. <b>Answer per your own institution's list</b>                                                                                                                        |
| Pharmacist's disaster role         | Maintain and rotate <b>disaster drug stock and antidote kits</b> (atropine–pralidoxime, potassium iodide, naloxone, antivenom, tetanus toxoid), bulk-prepare fluids and analgesics, control narcotic documentation, advise on dosing in renal failure/crush injury, manage the cold chain                                                                                   |

# 11. IN-SERVICE EDUCATION, BODY MECHANICS AND FATIGUE FACTORS

The PDF covers all three well. Only the quantitative anchors are missing.

- **Orientation** = initial familiarisation with policies and the environment (**AORN recommends a mentor / buddy system**). **In-service education** = ongoing, employer-provided, job-specific — perioperative departments hold it **monthly**. **Continuing education** = professional development beyond the employer.
- **Mandatory recurring training**: fire safety with **drills at least twice a year (unannounced)**, disaster drill annually, infection control and hand hygiene, biomedical waste, needle-stick protocol, radiation safety, laser safety, electrical safety, hazard communication, **BLS recertification every 2 years**, ergonomics, and **competency validation for every new device before use**.
- **Body mechanics**: lift by **bending the knees and hips with the back straight**, position your body under the load, hold it close, feet apart, **straighten the legs so the lower-limb muscles do the work, pivot with the feet — never twist the spine**, push rather than pull, lift in unison on a count.
- **NIOSH maximum recommended load = 51 lb (23 kg); patient-handling limit 35 lb (16 kg)**. OSHA (2002) requires facilities to reduce ergonomic injury; **lift teams** are cost-effective because back injury is the commonest occupational injury in health care.
- **Fatigue**: AORN recommends **≤ 12 consecutive hours** of direct patient care, **≤ 60 h/week including on-call**, and **8 h of uninterrupted sleep**. Being awake **17-19 hours** impairs performance about as much as a blood alcohol of 0.05 %. **Working with impaired judgement is an ethical breach, not just a personal risk**.
- **Other occupational hazards**: prolonged standing (varicose veins), repetitive strain in laparoscopy, **noise > 85 dB** from saws and drills, sharps injury, latex, and burnout.

## 12. RADIATION SAFETY

### 12.1 Units

| Quantity                    | SI unit            | Old unit     | Conversion                                                                         |
|-----------------------------|--------------------|--------------|------------------------------------------------------------------------------------|
| Absorbed dose               | Gray (Gy) = 1 J/kg | rad          | 1 Gy = 100 rad                                                                     |
| Equivalent / effective dose | Sievert (Sv)       | rem          | 1 Sv = 100 rem — the unit used for <b>all dose limits and personnel monitoring</b> |
| Activity                    | Becquerel (Bq)     | Curie (Ci)   | 1 Ci = 3.7 × 10 <sup>10</sup> Bq                                                   |
| Exposure                    | Coulomb/kg         | Roentgen (R) | Ionisation in air                                                                  |

### 12.2 Dose limits (ICRP / AERB)

| Category                         | Limit                                                                           |
|----------------------------------|---------------------------------------------------------------------------------|
| Occupational, whole body         | 20 mSv/year averaged over 5 years (100 mSv/5 y), maximum 50 mSv in any one year |
| Occupational, lens of the eye    | 20 mSv/year (ICRP 2011 revision; formerly 150 mSv)                              |
| Occupational, skin / extremities | 500 mSv/year                                                                    |
| Declared pregnancy               | ≤ 1 mSv to the fetus for the remainder of the pregnancy (0.5 mSv/month)         |
| Public                           | 1 mSv/year                                                                      |

## 12.3 Protection — three cardinal principles under ALARA

| Principle                               | Application                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Time</b>                          | Minimise screening time; <b>pulsed fluoroscopy, last-image hold, collimation</b> ; record total screening time                                                                                                                                                                                                                                                       |
| <b>2. Distance — INVERSE SQUARE LAW</b> | <b>Intensity <math>\propto 1/d^2</math></b> , so <b>doubling the distance gives <math>\frac{1}{4}</math> the exposure</b> and tripling gives $\frac{1}{3^2} = \frac{1}{9}$ . Stand <b><math>\geq 2</math> m (6 ft)</b> away or leave the room; in lateral views stand on the <b>image-intensifier side, away from the tube</b> (scatter is highest on the tube side) |
| <b>3. Shielding</b>                     | <b>Lead apron 0.25-0.5 mm lead equivalence</b> (0.5 mm attenuates $\approx 95-99\%$ at 100 kVp), <b>thyroid collar, lead glasses, lead gloves for holding cassettes, mobile lead screens</b> . Aprons are <b>hung, never folded</b> , and <b>screened annually for cracks</b>                                                                                        |
| <b>ALARA</b>                            | <b>As Low As Reasonably Achievable</b> — with ICRP's <b>justification, optimisation and dose limitation</b>                                                                                                                                                                                                                                                          |
| Monitoring                              | Dosimeter worn <b>OUTSIDE the lead apron at chest/collar level</b> (a second one <b>inside at waist level</b> in pregnancy). <b>TLD</b> is commonest; also film badge, OSL, electronic personal dosimeter. Records kept for life                                                                                                                                     |
| Personnel rules                         | Non-essential staff <b>leave the room or wear a lead apron</b> ; scrubbed staff <b>don the apron BEFORE scrubbing</b> ; <b>never hand-hold the cassette or patient</b> if avoidable; <b>pregnant staff avoid all radiation cases</b>                                                                                                                                 |
| Patient rules                           | <b>Ask every woman of child-bearing age if she is or could be pregnant</b> ; shield <b>gonads and thyroid</b> when the beam cannot be collimated; lowest diagnostic exposure                                                                                                                                                                                         |
| Radioactive implants                    | Handle with <b>long forceps, never fingers</b> ; lead pot in the room; limit and rotate staff time; exclude pregnant staff and children; <b>survey the room and linen with a Geiger-Müller counter</b> afterwards                                                                                                                                                    |

## 12.4 Biological effects

| Effect                                | Nature                                                                             | Examples                                                                                      |
|---------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Deterministic (non-stochastic)</b> | Has a <b>threshold</b> ; <b>severity</b> increases with dose                       | Skin erythema, epilation, <b>cataract</b> , sterility, marrow suppression, fetal malformation |
| <b>Stochastic</b>                     | <b>No threshold</b> ; <b>probability</b> increases with dose (linear-no-threshold) | <b>Carcinogenesis and heritable mutation</b>                                                  |

**Most radiosensitive: bone marrow and lymphoid tissue, gonads, intestinal crypts, skin basal layer, lens, and the fetus (most sensitive in weeks 2-8, organogenesis). Least sensitive: adult nerve, muscle and bone.**

## 13. INFECTION CONTROL

### 13.1 Standard vs transmission-based precautions

| Category                                                                                 | Applies to                                                                                                                                   | Requirements                                                                                                                                                                                                      | Examples                        |
|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Standard Precautions</b> (CDC 1996; absorbed the older <b>Universal Precautions</b> ) | <b>Every patient, every time</b> — all blood, body fluids, secretions and excretions <b>except sweat</b> , non-intact skin, mucous membranes | <b>Hand hygiene • gloves • gown • mask/eye protection when splash is likely • safe injection and sharps practice • respiratory hygiene • safe handling of equipment, linen and waste • environmental cleaning</b> | Universal application in the OR |

| Category        | Applies to                                               | Requirements                                                                                                                                    | Examples                                                                                      |
|-----------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Contact</b>  | Direct/indirect contact                                  | Standard + <b>gown and gloves for all contact</b> , dedicated equipment, single room preferred                                                  | <b>MRSA, VRE, CRE, C. difficile (soap and water; hypochlorite for surfaces), scabies, RSV</b> |
| <b>Droplet</b>  | Large droplets > 5 µm travelling ≈ <b>1-2 m (3-6 ft)</b> | Standard + <b>surgical mask within 1-2 m</b> , single room, mask the patient during transport                                                   | <b>Influenza, pertussis, mumps, rubella, N. meningitidis, diphtheria</b>                      |
| <b>Airborne</b> | Droplet nuclei ≤ 5 µm that stay suspended                | Standard + <b>fit-tested N95</b> , and an <b>Airborne Infection Isolation Room — NEGATIVE pressure with ≥ 12 air changes/hour</b> , door closed | <b>Pulmonary/laryngeal TB, measles, varicella</b>                                             |

### 13.2 PPE sequence — rote-learn the order

| Donning (put on)                                                                                                              | Doffing (take off)                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> Hand hygiene → <b>2</b> Gown → <b>3</b> Mask/respirator (fit-check) → <b>4</b> Goggles/face shield → <b>5</b> Gloves | <b>1</b> Gloves → <b>2</b> Goggles/face shield → <b>3</b> Gown → <b>4</b> Mask/respirator (remove <b>outside</b> the room, by the ties only) → <b>5</b> Hand hygiene |

The most contaminated item comes off first (gloves) and the respirator comes off last, because it protects you until you have left the room.

### 13.3 Sharps injury and post-exposure prophylaxis

| Item                                                   | Detail                                                                                                                                                                                                                                                    |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Transmission risk per percutaneous needle-stick</b> | <b>Hepatitis B 6-30 %</b> (up to 30 % if HBeAg-positive — the <b>highest</b> ) • <b>Hepatitis C ≈ 1.8 %</b> • <b>HIV ≈ 0.3 % (1 in 300)</b> ; mucous-membrane HIV exposure ≈ <b>0.09 %</b>                                                                |
| <b>Immediate first aid</b>                             | <b>Do NOT squeeze, suck, or apply bleach/alcohol/iodine into the wound. Wash with soap and running water</b> ; irrigate eye/mucosa with water or saline for <b>10-15 min</b> ; allow to bleed, dry and cover                                              |
| Then                                                   | <b>Report immediately</b> → assess the exposure → <b>baseline serology of both parties</b> → <b>start PEP</b> → counselling → follow-up at <b>6 weeks, 3 months, 6 months</b> → <b>document and file an incident report</b>                               |
| <b>HIV PEP</b>                                         | <b>As early as possible — ideally within 2 hours, never later than 72 hours</b> ; 3-drug regimen for <b>28 days</b>                                                                                                                                       |
| <b>Hepatitis B PEP</b>                                 | Unvaccinated or non-responder → <b>HBIG 0.06 mL/kg IM within 24 hours plus</b> start the vaccine course. A <b>vaccinated responder (anti-HBs ≥ 10 mIU/mL)</b> needs nothing. <b>HBV vaccination (0, 1, 6 months) is mandatory for health-care workers</b> |
| <b>Hepatitis C</b>                                     | <b>No vaccine and no PEP</b> — monitor and treat early infection with direct-acting antivirals                                                                                                                                                            |

| Item                 | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevention hierarchy | <b>Elimination/substitution (needleless systems) → engineering controls (retractable/sheathed needles, blunt suture needles, sharps container filled to <math>\frac{3}{4}</math>) → work-practice controls (never recap by hand — if unavoidable use the one-handed ‘scoop’; never pass a sharp hand-to-hand — use a neutral zone / hands-free technique with a magnetic pad; mount blades with a needle holder) → PPE (double gloving cuts inner-glove perforation by 70–80 %, face shield)</b> |

### 13.4 Biomedical waste segregation — BMW Rules 2016 (India)

| Colour                                                  | Contents                                                                                                                                                                                                                                                                                              | Treatment / disposal                                                                                                                                                                                              |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>YELLOW</b>                                           | <b>Human and animal anatomical waste</b> , soiled waste (dressings, cotton, blood-soaked items, plaster casts), <b>expired and discarded medicines</b> , chemical waste, contaminated linen, microbiology and laboratory waste, and <b>cytotoxic waste in a separate yellow ‘cytotoxic’ container</b> | <b>Incineration / plasma pyrolysis / deep burial</b> (deep burial only in towns < 500 000). <b>Cytotoxic waste → incineration at <math>\geq 1200\text{ }^{\circ}\text{C}</math></b> or return to the manufacturer |
| <b>RED</b>                                              | <b>Contaminated recyclable plastics</b> — IV tubing and sets, catheters, urine bags, <b>syringes without needles</b> , gloves                                                                                                                                                                         | <b>Autoclave / microwave / hydroclave, then shred → registered recycler</b> . No chemical pre-treatment, no landfilling                                                                                           |
| <b>WHITE</b><br>(translucent, puncture- and leak-proof) | <b>Waste sharps including metals</b> — needles, <b>syringes with fixed needles</b> , scalpels, blades                                                                                                                                                                                                 | <b>Autoclave or dry-heat sterilise, then shred/mutilate; final disposal by encapsulation in a concrete sharps pit</b> or to an iron foundry                                                                       |
| <b>BLUE</b>                                             | <b>Glassware</b> (broken/discarded contaminated glass, medicine vials and ampoules) and <b>metallic body implants</b>                                                                                                                                                                                 | <b>Disinfection (1 % hypochlorite) or autoclaving, then recycling</b>                                                                                                                                             |

- **Golden rule: segregate at the point of generation, by the person generating the waste.** Bags filled to  $\frac{3}{4}$ , tied (not stapled), labelled with the **biohazard/cytotoxic symbol**, stored  $\leq 48$  hours.
- **Never recap, bend or break needles by hand** — use a needle-tip cutter at the point of use; the needle goes to the **white** container.
- **Glass vials contaminated with cytotoxics go to YELLOW, not blue** — a classic trick question.
- **Mercury spill: do not vacuum**; collect with a syringe/adhesive tape or a spill kit into a sealed container of water and dispose as hazardous waste.
- Requirements: annual report to the Pollution Control Board, waste register, **annual health check-up and immunisation (hepatitis B, tetanus) of waste handlers**, PPE, and reporting of any major accident within 24 h. **Radioactive waste is governed separately by the AERB** (decay-in-storage).

## 14. CHEMICAL HAZARDS

### 14.1 Occupational exposure limits worth memorising

| Agent                | Limit                   | Effects and control                                                                                                                                                    |
|----------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nitrous oxide</b> | <b>NIOSH REL 25 ppm</b> | Megaloblastic change, neuropathy, <b>↑ spontaneous abortion and reduced fertility</b> . Control: <b>scavenging system, no leaks, tight masks, adequate air changes</b> |

| Agent                                    | Limit                                                  | Effects and control                                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Halogenated volatiles</b>             | <b>NIOSH REL 2 ppm</b> (0.5 ppm with N <sub>2</sub> O) | Headache, fatigue, <b>impaired mental performance, audiovisual ability and manual dexterity</b> ; hepatotoxicity; teratogenic concern                                                                                    |
| <b>Ethylene oxide</b>                    | <b>OSHA PEL 1 ppm</b> ; excursion 5 ppm/15 min         | Irritant, <b>mutagen, carcinogen, flammable/explosive</b> . Dedicated ventilated room, gas monitors, aeration cabinet, personal badges                                                                                   |
| <b>Formaldehyde</b>                      | <b>OSHA PEL 0.75 ppm</b> ; <b>STEL 2 ppm</b>           | Irritation, asthma, dermatitis, <b>Group 1 carcinogen</b> — why OT fumigation is abandoned. Handle specimens in a <b>ventilated hood</b>                                                                                 |
| <b>Glutaraldehyde</b>                    | <b>ACGIH ceiling 0.05 ppm</b>                          | <b>Occupational asthma, rhinitis, conjunctivitis, contact dermatitis, epistaxis</b> . Use a <b>covered basin with local exhaust, nitrile or butyl gloves (NOT latex — it permeates), goggles and apron</b>               |
| <b>Methyl methacrylate</b> (bone cement) | <b>OSHA PEL 100 ppm</b>                                | Irritant, dermatitis, headache; and in the patient <b>cement implantation syndrome — hypotension, hypoxia, arrhythmia, arrest during cementing. Mix in a closed vacuum system with local exhaust</b>                     |
| <b>Noise</b>                             | OSHA action level <b>85 dB (8-h TWA)</b>               | Saws, drills, suction, alarms — hearing protection; also impairs communication                                                                                                                                           |
| <b>Compressed gases</b>                  | —                                                      | <b>Secure every cylinder upright; never drop one or let cylinders strike each other; valve cap on for transport; never oil or grease an oxygen valve; segregate full and empty.</b> A falling cylinder becomes a missile |

## 14.2 Hazard communication, cytotoxics and spills

- **Hazard Communication / ‘Right to Know’**: every hazardous chemical needs a **Safety Data Sheet (SDS, formerly MSDS) in 16 standardised sections**, accessible to all staff at all times, plus **GHS labelling** with the signal word **‘Danger’** or **‘Warning’** and the nine pictograms.
- **Hierarchy of controls, in order**: elimination → substitution → engineering controls (ventilation, closed systems, scavenging) → administrative / work-practice controls → PPE (the **LAST and weakest line**). The ranking itself is commonly asked.
- **Cytotoxic handling**: prepare in a **Class II Type B biological safety cabinet or containment isolator in a negatively-pressured, externally-vented room**; use **closed-system transfer devices** and Luer-lock syringes; PPE = **two pairs of chemotherapy-tested gloves (changed every 30-60 min), an impermeable back-closing gown, eye protection and a respirator**; **never crush or split cytotoxic tablets**; **pregnant and breast-feeding staff must not handle them**; dispose in the **yellow cytotoxic stream**; keep a **spill kit**.
- **Chemical spill**: **evacuate and restrict access → identify the agent from the SDS → don PPE → contain and absorb from the outside inwards → neutralise → bag as hazardous waste → ventilate → report and document**. For a **skin or eye splash**, **irrigate at the eyewash station or shower for a minimum of 15 minutes**.
- **Waste anaesthetic gas control** (the PDF's topic, with the mechanism added): an efficient **scavenger system**, a leak-free circuit with well-fitting masks and connectors, **air-conditioning with HEPA filters that do not recirculate air, anaesthetic gas leak detectors**, and the same ventilation standard extended to the **PACU** because it adjoins the surgical suite.

## 15. POST-OPERATIVE PATIENT CARE

### WHY THIS SECTION IS KEPT

Your syllabus heading is “Preparation of Patient for Operation, **Pre & Post Operative Patient Care**”, but Chapter 1 of the PDF ends at the PACU hand-over and only reproduces a blank post-operative form. Everything below is new material.

It has been trimmed to the **examinable core**: PACU and scoring systems, immediate complications, the fever timeline, wound complications and drains. Fluid regimens, nutrition and analgesic pharmacology have been cut as off-syllabus.

## 15.1 Phases of recovery and the PACU

| Phase                          | Location                                            | Focus                                                                                    |
|--------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Phase I — immediate</b>     | PACU / recovery room, <b>1 nurse : 1-2 patients</b> | Emergence; airway, breathing, circulation, consciousness, temperature, pain, haemorrhage |
| <b>Phase II — intermediate</b> | Step-down or ambulatory unit, ward                  | Preparation for discharge home or to the ward; oral intake, ambulation, teaching         |
| <b>Phase III — extended</b>    | Ward, home                                          | Rehabilitation, wound care, complication surveillance                                    |

- **Hand-over must include:** procedure and findings, **type of anaesthesia and reversal**, airway status, **allergies**, co-morbidity, intra-operative events and vital-sign trends, **blood loss, fluids and blood products given, urine output**, drains/catheters/packs, **analgesia and antiemetics given with times**, laterality, post-operative orders and whom to call. **The anaesthetist does not leave until the PACU nurse accepts the patient and the hand-over is documented.**
- **Monitoring:** **vital signs every 5-15 min** initially, then 15-minutely until stable — **RR and pattern, SpO<sub>2</sub>, BP, pulse, temperature, ECG, pain score, level of consciousness**, plus **hourly urine and drain output**.
- **Position:** **lateral or head-up 15-30°** (keeps the tongue forward, lets secretions drain); **supine flat** after spinal anaesthesia or with hypotension. **Side rails up, oxygen on, suction and an Ambu bag at the bedside, and the patient is never left unattended.**
- **Check:** dressing for ooze, drain patency and character, IV site, **peripheral pulses and capillary refill distal to any cast or tourniquet, ability to move limbs and any paraesthesia** (rules out a positioning injury and assesses spinal regression), and bladder distension.
- **Warm actively** — forced-air blanket, warmed fluids; aim for core temperature **> 36 °C** before discharge from PACU.

## 15.2 Aldrete Score — memorise this table

| Parameter                | 2                                             | 1                                                    | 0                               |
|--------------------------|-----------------------------------------------|------------------------------------------------------|---------------------------------|
| <b>Activity</b>          | Moves <b>all four limbs</b> on command        | Moves <b>two</b> extremities                         | Unable to move                  |
| <b>Respiration</b>       | Breathes deeply and <b>coughs freely</b>      | Dyspnoeic or shallow                                 | Apnoeic / ventilated            |
| <b>Circulation (BP)</b>  | <b>Within ± 20 %</b> of pre-anaesthetic level | ± 20-49 %                                            | ± ≥ 50 %                        |
| <b>Consciousness</b>     | <b>Fully awake</b>                            | Rousable on calling                                  | Not responding                  |
| <b>Oxygen saturation</b> | <b>SpO<sub>2</sub> &gt; 92 % on room air</b>  | Needs O <sub>2</sub> to keep SpO <sub>2</sub> > 90 % | < 90 % even with O <sub>2</sub> |

## SCORING — EXAM POINTS

**Maximum 10; discharge from PACU requires  $\geq 9$**  with no parameter scoring 0, plus stable vitals, controlled pain and PONV, no active bleeding and normothermia. Mnemonic **A-R-C-C-O**.

The **original 1970 Aldrete score used skin COLOUR**; the **modified score replaced colour with pulse-oximetry SpO<sub>2</sub>**.

**Pain and nausea are NOT in the Aldrete score** — they belong to the **PADSS**. This inversion is a classic MCQ.

**PADSS** (Post-Anaesthesia Discharge Scoring System, for discharge **home** after day surgery) scores 0–2 each for **vital signs • ambulation • nausea/vomiting • pain • surgical bleeding**;  **$\geq 9$  required**, plus a **responsible escort, written instructions and tolerance of oral fluids**.

Counsel: **no driving, machinery, legal documents, alcohol or sole care of a child for at least 24 hours** after general anaesthesia or sedation.

After spinal/epidural anaesthesia also document **return of motor power, sensation and the sensory level**.

## 15.3 Immediate post-operative complications

| Complication                                                       | Cause                                                                                                                                                                     | Recognition                                                                                                                | Management                                                                                                                                                                                         |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Airway obstruction — the commonest immediate PACU emergency</b> | <b>Tongue falling back against the posterior pharynx</b> in the sedated supine patient; secretions, blood, laryngeal oedema                                               | <b>Snoring or stridor, see-saw (paradoxical) chest movement, tracheal tug, desaturation</b>                                | <b>Head tilt-chin lift / jaw thrust</b> , lateral position, suction, oral or nasal airway, 100 % O <sub>2</sub> ; re-intubate if needed                                                            |
| <b>Laryngospasm</b>                                                | Cord stimulation in light anaesthesia, secretions, blood                                                                                                                  | <b>Inspiratory stridor progressing to complete silence</b> , paradoxical effort                                            | Remove the stimulus, suction, <b>100 % O<sub>2</sub> with CPAP and jaw thrust</b> ; if persistent — propofol then <b>succinylcholine</b> and intubate                                              |
| <b>Hypoventilation</b>                                             | <b>Residual anaesthetic, opioids, incomplete reversal of neuromuscular blockade</b> , pain splinting, obesity, hypothermia                                                | Slow shallow breathing, $\uparrow$ EtCO <sub>2</sub> , drowsiness, pin-point pupils                                        | Stimulate, O <sub>2</sub> , support ventilation; <b>naloxone 40–80 <math>\mu</math>g increments</b> for opioids, <b>neostigmine/sugammadex</b> for relaxant, <b>flumazenil</b> for benzodiazepines |
| <b>Atelectasis — the commonest post-op pulmonary complication</b>  | Shallow breathing, retained secretions, upper-abdominal or thoracic incision                                                                                              | <b>Low-grade fever POD 1–2</b> , reduced basal breath sounds, tachypnoea                                                   | <b>Deep breathing and incentive spirometry (10 breaths hourly), splinted coughing, early ambulation, adequate analgesia, physiotherapy</b> — not antibiotics                                       |
| <b>Hypotension</b>                                                 | <b>Hypovolaemia (commonest — blood loss, third-space loss)</b> , residual anaesthetic, sympathetic block from spinal/epidural, arrhythmia, ischaemia, sepsis, anaphylaxis | <b>Check the surgical site and drains for bleeding FIRST</b> ; tachycardia and a narrow pulse pressure precede hypotension | Head-down/legs up, O <sub>2</sub> , <b>crystalloid bolus 10–20 mL/kg</b> , vasopressor if needed, ECG, Hb, cross-matched blood; <b>surgical control is definitive</b>                              |
| <b>Hypertension</b>                                                | <b>Pain (commonest) and bladder distension</b> , hypoxia/hypercarbia, hypothermic shivering, omitted antihypertensives, fluid overload                                    | Raised BP with restlessness                                                                                                | <b>Treat the cause first — analgesia, empty the bladder, oxygenate, warm</b> — before giving antihypertensives                                                                                     |



| Complication                          | Cause                                                                                                                                        | Recognition                                                                                                                                                        | Management                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Delayed emergence</b>              | Residual drugs; <b>hypoglycaemia, hypoxia, hypercarbia, hyponatraemia, hypothermia (&lt; 33 °C prevents awakening)</b> ; stroke, seizure     | Failure to wake as expected                                                                                                                                        | <b>Check glucose, ABG, electrolytes and pupils;</b> reverse opioids and benzodiazepines; <b>think stroke if there are focal signs</b>                                          |
| <b>Emergence delirium / agitation</b> | Commonest in <b>children and the elderly</b> ; also hypoxia, pain, full bladder, ketamine, withdrawal                                        | Restlessness, disorientation, thrashing                                                                                                                            | <b>Always exclude hypoxia first,</b> then pain and a full bladder; reassure and orientate; avoid physical restraint                                                            |
| <b>Hypothermia and shivering</b>      | Cold OR, anaesthetic vasodilatation, cold fluids, large exposed cavity, elderly and neonates                                                 | Core < 36 °C; shivering <b>raises O<sub>2</sub> consumption up to 400 %</b> ; causes <b>coagulopathy, ↑ blood loss, ↑ SSI, arrhythmia, delayed drug metabolism</b> | <b>Forced-air warming, warmed fluids and irrigation, raised room temperature;</b> oxygen; <b>pethidine (meperidine) 12.5-25 mg IV</b> is the most effective drug for shivering |
| <b>Haemorrhage</b>                    | <b>Reactionary — within 24 h</b> (slipped ligature, clot dislodged as BP rises); <b>Secondary — POD 7-14 from infection eroding a vessel</b> | Tachycardia, narrow pulse pressure, restlessness, cool clammy skin, oliguria, brisk drain output                                                                   | Two large-bore cannulae, crystalloid, <b>cross-matched blood / massive transfusion protocol</b> , tranexamic acid, keep warm, <b>return to theatre</b>                         |

## 15.4 Post-operative nausea and vomiting (PONV) — in brief

- Incidence **20-30 %**, up to **80 %** in high-risk patients; the commonest reason for unplanned admission after day surgery.
- Apfel score — 1 point each for: female sex • non-smoker • previous PONV or motion sickness • expected post-operative opioid use.** Risk for 0/1/2/3/4 factors  $\approx 10 / 20 / 40 / 60 / 80$  %. **Smoking is protective.**
- Other risks: volatile agents and **nitrous oxide**, long anaesthesia, and gynaecological, laparoscopic, middle-ear, squint and breast surgery.
- Drugs by receptor: 5-HT<sub>3</sub> antagonist — ondansetron 4-8 mg IV, given at the END of surgery (watch QT prolongation) • steroid — dexamethasone 4-8 mg at induction • D<sub>2</sub> antagonist — metoclopramide (extrapyramidal effects; avoid in obstruction) • NK<sub>1</sub> — aprepitant • antihistamine — cyclizine • anticholinergic — hyoscine patch • propofol TIVA is itself antiemetic.**
- Strategy: 1-2 risk factors → two agents from DIFFERENT classes; ≥ 3 → three or four agents plus propofol TIVA, avoiding nitrous oxide and sparing opioids.** For breakthrough PONV use a class **not already given.**

## 15.5 Post-operative fever — the “W's” timeline (very high yield)

| Onset                        | “W”                               | Cause                                                                                                                             | Action                                                                                          |
|------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>During or within 24 h</b> | <b>What we did / Wonder</b> drugs | Inflammatory (cytokine) response to surgery, <b>drug or transfusion reaction, MALIGNANT HYPERTHERMIA</b> , pre-existing infection | <b>Usually NOT infective.</b> Rigidity + ↑ EtCO <sub>2</sub> → dantrolene; stop any transfusion |
| <b>POD 1-2</b>               | <b>Wind</b>                       | <b>ATELECTASIS — the commonest cause of early post-op fever;</b> aspiration, early pneumonia                                      | Incentive spirometry, deep breathing, ambulation, analgesia, physiotherapy                      |

| Onset             | “W”          | Cause                                                                                                  | Action                                                                                            |
|-------------------|--------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>POD 3-5</b>    | Water        | <b>Urinary tract infection</b> (catheter-associated)                                                   | Urinalysis and culture; <b>remove or change the catheter</b> ; antibiotics                        |
| <b>POD 4-6</b>    | Walking      | <b>Deep-vein thrombosis / pulmonary embolism</b>                                                       | Calf pain and unilateral swelling, or sudden dyspnoea and hypoxia → Doppler / CTPA; anticoagulate |
| <b>POD 5-7</b>    | Wound        | <b>Surgical site infection</b> ; also <b>anastomotic leak</b> (with tachycardia and ileus)             | Inspect → <b>open, drain and culture</b> ; suspect a leak if the patient is ‘not doing well’      |
| <b>&gt; POD 7</b> | Wonder drugs | <b>Drug fever</b> , abscess/collection, C. difficile colitis, IV-site thrombophlebitis, line infection | Review the drug chart; culture blood, urine and line tips; CT for a collection                    |

### **MNEMONIC — Wind, Water, Walking, Wound, Wonder drugs**

Time order: **Wind (1-2) → Water (3-5) → Walking (4-6) → Wound (5-7) → Wonder drugs (> 7).**

**Fever in the first 24 hours is usually NOT infection. Fever after POD 5 with wound signs is infection until proved otherwise.**

## **15.6 Wound healing and wound complications**

| Phase                              | Timing                    | Events                                                                                           | Dominant cell                                                       |
|------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Haemostasis</b>                 | Minutes-hours             | Vasoconstriction, platelet plug, fibrin clot                                                     | Platelets                                                           |
| <b>Inflammatory (Iag)</b>          | <b>Day 0-4/5</b>          | Vasodilatation, exudate, phagocytosis, debridement; wound held only by clot and sutures          | <b>Neutrophils (24-48 h) → MACROPHAGES (48-72 h — the key cell)</b> |
| <b>Proliferative (fibroplasia)</b> | **Day 4-21                | <b>Collagen III</b> deposition, angiogenesis, granulation tissue, epithelialisation, contraction | <b>Fibroblasts</b>                                                  |
| <b>Maturation / remodelling</b>    | <b>Day 21 → 1-2 years</b> | <b>Collagen III replaced by collagen I</b> , cross-linking, scar pales and flattens              | Fibroblasts, collagenase                                            |

- **Tensile strength:** ≈ 5-10 % at 1 week, 30-50 % at 1 month, maximum only 70-80 % of normal skin — which is why dehiscence peaks in the first week and lifting is restricted for six weeks.
- **Types of healing:** **primary (first) intention** — apposed edges, fine scar; **secondary intention** — left open, heals by granulation and contraction (contaminated wounds, abscess cavities); **tertiary / delayed primary closure** — left open 3-5 days then closed.
- **Factors delaying healing** — local: **infection, ischaemia, haematoma, foreign body, tension, dead space, movement, radiation**; systemic: **age, diabetes, malnutrition (protein, vitamin C, vitamin A, zinc), anaemia and hypoxia, obesity, smoking, steroids and immunosuppressants, chemotherapy, jaundice, uraemia, malignancy.**
- **Dressings:** the first dressing over a primarily closed wound is left undisturbed **24-48 h** (epithelialisation is complete and it acts as a bacterial barrier); thereafter change with **aseptic technique**, inspect and document; change immediately if soaked (**strike-through**).
- **Suture removal:** face and neck **3-5 days**; scalp **7-10**; trunk/abdomen **7-10**; upper and lower limb **10-14**; over joints, back, palms and soles **14 days**.

| Complication                        | Timing and features                                                                                                                                                                                                            | Management                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Haematoma</b>                    | 24–48 h; swelling, discoloration, pain. <b>A neck haematoma after thyroidectomy is an AIRWAY EMERGENCY</b>                                                                                                                     | Small → observe; expanding or compressive → <b>open and evacuate</b> (at the bedside for the neck)                                                                                                                                                                                                                                      |
| <b>Seroma</b>                       | Day 3–10, especially after <b>mastectomy, axillary or groin dissection, mesh hernia repair</b> ; fluctuant, painless                                                                                                           | Aseptic aspiration, compression, drains                                                                                                                                                                                                                                                                                                 |
| <b>Surgical site infection</b>      | <b>POD 5–7</b> ; pain, erythema, warmth, induration, purulent discharge, fever. <b>Spreading cellulitis or crepitus on POD 1–2 → suspect streptococcal or clostridial necrotising infection — a surgical emergency</b>         | <b>Open, drain, culture, debride, pack for secondary healing</b> , culture-guided antibiotics — ‘the treatment of pus is drainage’                                                                                                                                                                                                      |
| <b>Dehiscence</b>                   | <b>POD 5–10 (classically day 7–8)</b> ; the herald sign is a <b>sudden serosanguineous ‘salmon-pink’ discharge</b> with a palpable gap, often after a cough                                                                    | Abdominal binder, avoid straining, <b>inform the surgeon → usually re-suturing in theatre</b>                                                                                                                                                                                                                                           |
| <b>Evisceration (burst abdomen)</b> | Protrusion of viscera through the wound                                                                                                                                                                                        | <b>EMERGENCY: stay with the patient; supine, low Fowler's with knees flexed; COVER the viscera with sterile gauze soaked in WARM sterile normal saline — never dry gauze and never push the bowel back; keep nil by mouth; IV access and fluids; monitor for shock; analgesia; notify the surgeon and prepare for immediate theatre</b> |
| <b>Incisional hernia</b>            | Weeks to years later — the late consequence of subclinical dehiscence or infection                                                                                                                                             | Elective mesh repair                                                                                                                                                                                                                                                                                                                    |
| <b>Hypertrophic scar vs keloid</b>  | <b>Hypertrophic</b> stays <b>within</b> the original margins and tends to regress; <b>keloid extends BEYOND</b> the margins, does not regress and recurs after excision (commoner in darker skin, sternum, shoulder, ear lobe) | Silicone sheeting, pressure, intralesional steroid                                                                                                                                                                                                                                                                                      |

## 15.7 Drains, and the rest of the routine

| Drain                       | Type                                         | Key point                                                                                      |
|-----------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Penrose / corrugated</b> | <b>Passive, open</b>                         | Gravity or capillary action; higher infection risk                                             |
| <b>Jackson-Pratt (bulb)</b> | <b>Closed, active</b>                        | <b>Re-compress the bulb after emptying</b> — a fully expanded bulb means suction has been lost |
| <b>Hemovac / Redivac</b>    | <b>Closed, active (spring / high vacuum)</b> | Orthopaedic, breast and neck surgery                                                           |

| Drain                            | Type                           | Key point                                                                                                                                                                                                                                       |
|----------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chest (intercostal) drain</b> | <b>Closed, underwater seal</b> | <b>Keep the bottle below chest level; never clamp a bubbling drain; 'swinging' = patent, continuous bubbling = air leak.</b> If disconnected → place the end in sterile water; if it falls out → occlusive dressing taped on <b>three sides</b> |
| <b>Nasogastric tube</b>          | Decompression / feeding        | Confirm position by <b>pH &lt; 5.5 on aspirate or X-ray</b> — the auscultation 'whoosh' test is unreliable and unsafe                                                                                                                           |
| <b>Urinary catheter</b>          | Closed drainage                | <b>Closed system, bag below bladder level and off the floor, no dependent loops, remove within 24 h. CAUTI is the commonest healthcare-associated infection</b>                                                                                 |

- **Drain care:** secure against traction, keep patent, chart **volume, colour and character** each shift, aseptic site dressing, and **never advance a drain back in. Brisk bright-red output → haemorrhage; faecal or bilious output → anastomotic leak; cloudy or foul → infection** — report immediately.
- **Bowel function returns in the order: small intestine ≈ 24 h → stomach 24-48 h → COLON 48-72 h (last).** Resolution is judged by the **passage of flatus or stool**, not bowel sounds. **Ileus** — absent bowel sounds and generalised distension; **mechanical obstruction** — colicky pain with high-pitched tinkling sounds.
- **Urine output must exceed 0.5 mL/kg/h (≈ 30 mL/h).** Post-operative oliguria is **most often hypovolaemia; urinary retention** (commonest after anorectal, hernia, pelvic and spinal-anaesthetic surgery) shows as suprapubic fullness, restlessness and overflow dribbling → privacy, upright position, warm compress, analgesia, then **in-and-out catheterisation**.
- **DVT peaks POD 5-10** and is often silent; **PE** presents as sudden dyspnoea, pleuritic pain, tachycardia and hypoxia, and is a leading cause of **preventable** post-operative death. Prevention = **early ambulation, ankle exercises, TED stockings, intermittent pneumatic compression, LMWH, hydration and adequate analgesia so the patient can move.**
- **Respiratory bundle:** semi-Fowler's position, **incentive spirometry 10 breaths hourly, splinted coughing**, adequate analgesia, **ambulation within 24 h**, oral hygiene, humidified oxygen.
- **Red flags to teach the patient before discharge:** fever, increasing pain, spreading redness, purulent or foul discharge, wound gaping, bleeding, calf pain or swelling, chest pain or breathlessness, persistent vomiting, inability to pass urine.

## MEDICATION SAFETY IN THE OR — the pharmacist's angle on this unit

Extend the PDF's **five rights** to the **eight rights**: right **patient, drug, dose, route, time**, plus right **documentation, indication and response**.

**Every medication on the sterile field is labelled immediately** with name, strength and concentration; **unlabelled medications are ALWAYS discarded**; **all vials and syringes stay in the room until the end of the case**; labelled medications are **shown to the relief person** at hand-over; **calculations are independently double-checked by a second licensed professional**.

**High-alert drugs in the OR**: concentrated **potassium chloride**, insulin, heparin, **neuromuscular blockers** ('paralysing agent — warning' label), concentrated adrenaline, local anaesthetics, oxytocin, and **look-alike/sound-alike pairs** (ephedrine/epinephrine).

**Constraints recommended by AORN**: minimise the number of calculations required, stock **dose-specific ready-to-use concentrations**, standardise concentrations, use pre-filled syringes, and assess competency regularly.

Also the pharmacist's OT duties: crash-cart stock, expiry and seal checks, **cold-chain logs**, **narcotic accounting (NDPS register)**, sterile dilution of injectables, **ADR reporting (pharmacovigilance)** and in-service education on new drugs and devices.

## 16. THE NUMBERS SHEET

| Topic                                | Number                                                                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fasting                              | <b>2 h clear fluids • 4 h breast milk • 6 h formula/light meal • 8 h fatty meal</b> ; aspiration risk if gastric volume > <b>25 mL</b> and pH < <b>2.5</b>                   |
| Antibiotic prophylaxis               | <b>Within 60 min of incision</b> ; re-dose after <b>2 half-lives</b> or > <b>1500 mL</b> blood loss; <b>stop within 24 h</b>                                                 |
| Identification                       | ≥ <b>2 identifiers</b> ; WHO checklist = <b>3 phases, 19 items</b> ; Universal Protocol <b>July 2004</b> ; AORN position addition <b>March 2005</b>                          |
| Safety strap                         | Mid-thighs, ≈ <b>3 inches (7.5 cm) above the knees</b> ; arms abducted ≤ <b>90°</b> ; Trendelenburg <b>30-45°</b>                                                            |
| Sterile field                        | Wrapper edge unsterile <b>2.5 cm (1 in)</b> • unsterile person keeps <b>30 cm (1 ft)</b> away • gown sleeves sterile to <b>5 cm above the elbow</b>                          |
| Scrub                                | First scrub of the day <b>3-5 min</b> ; double gloving cuts perforation <b>70-80 %</b>                                                                                       |
| OR environment                       | <b>20-23 °C • RH 20-60 % • ≥ 15 (ideally 20-25) air changes/h • POSITIVE pressure • HEPA 99.97 % at 0.3 µm • laminar flow 300-600 ACH • noise &lt; 45 dB</b>                 |
| Wound class SSI risk                 | <b>Clean &lt; 2 % • clean-contaminated 3-10 % • contaminated 15-20 % • dirty 30-40 %</b> ; SSI window <b>30 days (90 with an implant)</b>                                    |
| Autoclave                            | <b>121 °C / 15 psi / 15-20 min • 134 °C / 30 psi / 3-3.5 min • prevacuum 132-135 °C / 3-4 min • flash 132 °C / 3 min unwrapped (10 min lumened) • prions 134 °C / 18 min</b> |
| Hot air oven                         | <b>160 °C / 2 h • 170 °C / 1 h • 180 °C / 30 min</b>                                                                                                                         |
| Bowie-Dick                           | <b>134 °C for 3.5 min in an EMPTY prevacuum chamber, daily</b> — tests <b>air removal</b>                                                                                    |
| Ethylene oxide                       | <b>450-1200 mg/L • 37-63 °C • RH 40-80 % • 1-6 h</b> ; aeration <b>8-12 h at 50-60 °C</b> or <b>7 days</b> at room temperature                                               |
| H <sub>2</sub> O <sub>2</sub> plasma | <b>58 % H<sub>2</sub>O<sub>2</sub> • 45-50 °C • ≈ 1 h • no cellulose, no liquids</b>                                                                                         |
| Liquid chemicals                     | <b>Glutaraldehyde 2 % — 10 h sterilization / 20-45 min HLD • OPA 0.55 % — 12 min • peracetic acid — 12 min at 50-56 °C • H<sub>2</sub>O<sub>2</sub> 7.5 % — 30 min HLD</b>   |
| Sterility                            | <b>SAL 10<sup>-6</sup> • γ-radiation 25 kGy • filter 0.22 µm • shelf-life is event-related</b>                                                                               |

| Topic                  | Number                                                                                                                                                                                                                                                                                                              |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Biological indicators  | <b>Steam &amp; plasma → Geobacillus stearothermophilus • dry heat &amp; EO → Bacillus atrophaeus • radiation → Bacillus pumilus</b>                                                                                                                                                                                 |
| Electrosurgery         | <b>300 kHz - 3 MHz</b> ; no nerve stimulation above <b>~100 kHz</b> ; ultrasonic scalpel <b>55.5 kHz</b> ; LigaSure seals <b>7 mm</b> ; smoke — 1 g tissue $\approx$ <b>3-6 cigarettes</b> , evacuator within <b>1-2 in</b> , <b>ULPA 99.999 % at 0.1 <math>\mu</math>m</b>                                         |
| Tourniquet             | <b>Upper limb 250-300 mmHg / <math>\leq</math> 60 min • lower limb 300-350 mmHg / <math>\leq</math> 90-120 min</b> ; overlap <b>3-6 in</b> ; deflate <b>10-15 min</b> if longer                                                                                                                                     |
| CPR                    | <b>100-120/min • 5-6 cm • 30:2 • biphasic 120-200 J • adrenaline 1 mg q3-5 min • amiodarone 300 mg</b>                                                                                                                                                                                                              |
| Anaphylaxis            | <b>Adrenaline 0.5 mg IM (1:1000)</b> , repeat q5 min; IV <b>50 <math>\mu</math>g</b> boluses; fluid <b>20 mL/kg</b>                                                                                                                                                                                                 |
| Malignant hyperthermia | <b>Dantrolene 2.5 mg/kg up to 10 (-30) mg/kg • O<sub>2</sub> 10 L/min • cool to 38 °C • urine &gt; 1-2 mL/kg/h • RYR1, chromosome 19</b>                                                                                                                                                                            |
| Radiation              | <b>1 Gy = 100 rad • 1 Sv = 100 rem</b> ; occupational <b>20 mSv/y (max 50 in one year)</b> ; public <b>1 mSv/y</b> ; eye <b>20 mSv/y</b> ; extremities <b>500 mSv/y</b> ; fetus <b>1 mSv</b> (0.5 mSv/month); apron <b>0.25-0.5 mm Pb</b> ; stand $\geq$ <b>2 m</b> ; doubling distance $\rightarrow$ $\frac{1}{4}$ |
| Needle-stick           | <b>HBV 6-30 % • HCV 1.8 % • HIV 0.3 %</b> ; HIV PEP within <b>2 h (max 72 h) for 28 days</b> ; HBIG <b>0.06 mL/kg within 24 h</b>                                                                                                                                                                                   |
| Chemical limits        | <b>EO 1 ppm • formaldehyde 0.75 ppm • glutaraldehyde 0.05 ppm ceiling • N<sub>2</sub>O 25 ppm • volatiles 2 ppm • methyl methacrylate 100 ppm • noise 85 dB</b>                                                                                                                                                     |
| Lifting                | <b>NIOSH 51 lb (23 kg)</b> ; patient handling <b>35 lb (16 kg)</b>                                                                                                                                                                                                                                                  |
| Waste                  | <b>Yellow</b> anatomical/soiled/expired drugs/cytotoxic • <b>Red</b> recyclable plastics • <b>White</b> sharps • <b>Blue</b> glass and metallic implants; bags to $\frac{3}{4}$ ; cytotoxic incineration $\geq$ <b>1200 °C</b> ; storage $\leq$ <b>48 h</b>                                                         |
| Fatigue & drills       | $\leq$ <b>12 consecutive hours • <math>\leq</math> 60 h/week • 8 h sleep</b> ; fire drills $\geq$ <b>2/year</b> ; BLS recertification <b>every 2 years</b>                                                                                                                                                          |
| Recovery scores        | <b>Aldrete max 10, discharge <math>\geq</math> 9 (A-R-C-C-O) • PADSS <math>\geq</math> 9 • Apfel 4 factors <math>\rightarrow</math> 10/20/40/60/80 %</b>                                                                                                                                                            |
| Post-op timeline       | <b>Fever: Wind 1-2 • Water 3-5 • Walking 4-6 • Wound 5-7 • Drugs &gt; 7 • dehiscence POD 5-10 • secondary haemorrhage POD 7-14 • DVT POD 5-10</b>                                                                                                                                                                   |
| Wound healing          | Inflammatory <b>0-4 d</b> • proliferative <b>4-21 d</b> • remodelling <b>21 d-1 y</b> ; tensile strength <b>5-10 % at 1 wk, 30-50 % at 1 month, max 70-80 %</b> ; sutures out <b>face 3-5, trunk 7-10, limbs 10-14, joints 14 d</b>                                                                                 |
| Bowel & urine          | <b>Small bowel 24 h <math>\rightarrow</math> stomach 24-48 h <math>\rightarrow</math> colon 48-72 h</b> ; urine output <b>&gt; 0.5 mL/kg/h</b>                                                                                                                                                                      |

## 17. MCQ BANK — 77 QUESTIONS, SYLLABUS-WEIGHTED

### ABOUT THESE QUESTIONS

**77 questions**, every one mapped to a listed syllabus sub-point. Questions on off-syllabus topics (ASA grading, premedication pharmacology, analgesic choice, fluid regimens) have been removed from the earlier draft. They are **modelled on the pattern and difficulty of previous-year questions** in Indian pharmacy, nursing and OT-technology papers, and **written from the standard sources** — AORN Guidelines, Berry & Kohn's Operating Room Technique, CDC/HICPAC guidelines, WHO, ICRP/AERB, OSHA/NIOSH, BMW Rules 2016 and Bailey & Love. They are **not verbatim reproductions** of any single paper. Weighting, matched to where the marks fall: **sterilization 14 • post-operative care 13 • environment/asepsis/SSI 10 • electrosurgery & equipment 9 • pre-op preparation, identification, records & counting 9 • positioning 8 • emergencies 7 • radiation, chemical hazards & waste 7.**

## A — Pre-Operative Preparation, Identification and Records

### Q1. The minimum fasting period for clear liquids before elective surgery in a healthy adult is

- (a) 1 hour (b) 2 hours (c) 4 hours (d) 6 hours

**Ans: (b) 2 hours** — Remember **2-4-6-8**: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h. The purpose is to prevent **aspiration (Mendelson's syndrome)**; prolonged fasting is harmful and does not reduce gastric volume.

### Q2. The preferred method of pre-operative hair removal is

- (a) Shaving with a razor the night before (b) Shaving with a razor in the OR (c) Electric clippers immediately before surgery (d) Hair is always left intact

**Ans: (c) Electric clippers immediately before surgery** — Ideally hair is **not removed**; if it interferes, use **clippers immediately before surgery, outside the OR**. Razor shaving — especially the night before — creates micro-abrasions and carries the **highest** SSI risk.

### Q3. The WHO Surgical Safety Checklist consists of how many phases?

- (a) Two (b) Three (c) Four (d) Five

**Ans: (b) Three** — **Three: Sign In** (before induction), **Time Out** (after induction, before incision), **Sign Out** (before leaving the OR) — 19 items. It reduced mortality from 1.5 % to 0.8 %.

### Q4. The surgical “time out” must be performed

- (a) On admission to the ward (b) In the holding area (c) After induction but before skin incision (d) Immediately after wound closure

**Ans: (c) After induction but before skin incision** — The time out is the whole-team verbal pause **immediately before incision**. The **Universal Protocol** was mandated by the Joint Commission in **July 2004**; **AORN added the correct patient position in March 2005**.

### Q5. Which of the following is verified during the “Sign Out” phase?

- (a) Pulse oximeter functioning (b) Site marking (c) Correctness of instrument, sponge and needle counts (d) Difficult airway risk

**Ans: (c) Correctness of instrument, sponge and needle counts** — Sign Out covers the **name of the procedure recorded, correctness of counts, specimen labelling read aloud, equipment problems and key recovery concerns**. Pulse oximetry, site marking and airway risk belong to **Sign In**.

### Q6. A specimen for frozen section must be sent

- (a) In 10 % formalin (b) Fresh, in saline-moistened gauze (c) In absolute alcohol (d) In 2 % glutaraldehyde

**Ans: (b) Fresh, in saline-moistened gauze** — **Frozen sections go FRESH — never in formalin**, which prevents further processing and immunohistochemistry. Routine histopathology goes in **10 % neutral buffered formalin at  $\approx 10\times$  the specimen volume**; culture specimens go in a sterile container with no fixative.

### Q7. Consent for surgery on a 10-year-old child is governed by which section of the IPC?

- (a) Section 87 (b) Section 88 (c) Section 89 (d) Section 92

**Ans: (c) Section 89 — § 89** — acts done in good faith for a **child under 12** or a person of unsound mind, with the guardian's consent. **§ 90** — consent given under fear or misconception is invalid. **§ 92** — acts in good faith **without** consent in an emergency.

### Q8. A retained surgical sponge presenting years later as an abdominal mass is called a

- (a) Granuloma annulare (b) Gossypiboma (c) Pseudocyst (d) Phlegmon

**Ans: (b) Gossypiboma** — **Gossypiboma (textiloma)** — gauze encased in a foreign-body granuloma. The sponge is the **commonest retained item (50-70 %)** and this is a classic **res ipsa loquitur** claim. Risk factors: emergency surgery, **unexpected change of procedure** and high BMI.

**Q9. The legal responsibility for recording the surgical count rests with the**

- (a) Surgeon (b) Scrub person (c) Circulator (d) Anaesthetist

**Ans: (c) Circulator** — Both count **aloud together** as the scrub person touches each item, but **keeping the written record is the circulator's legal responsibility**, and the circulator initials every addition.

**B — Transfer and Positioning**

**Q10. The single most important measure to prevent a patient falling from the operating table is**

- (a) Side rails (b) A safety strap firmly applied across the mid-thighs (c) Padded armboards (d) Two staff at the bedside

**Ans: (b) A safety strap firmly applied across the mid-thighs** — The strap across the **mid-thighs, ≈ 3 inches (7.5 cm) above the knees**, firm but not tight enough to impair venous return. The patient is also **never left unattended** in the OR.

**Q11. The nerve most commonly injured in the lithotomy position is the**

- (a) Femoral nerve (b) Sciatic nerve (c) Common peroneal nerve (d) Obturator nerve

**Ans: (c) Common peroneal nerve** — The **common peroneal nerve** is compressed against the **fibular head** by the stirrup post → **foot drop** and loss of dorsal foot sensation. Lithotomy also risks **lower-limb compartment syndrome** if prolonged.

**Q12. The commonest peripheral nerve injury in the supine position involves the**

- (a) Radial nerve (b) Ulnar nerve (c) Median nerve (d) Axillary nerve

**Ans: (b) Ulnar nerve** — **Ulnar neuropathy** is the commonest peri-operative nerve injury overall. Prevent by abducting the arms **≤ 90°**, padding the elbows and placing the **palms up (supinated)**.

**Q13. All of the following occur in the Trendelenburg position EXCEPT**

- (a) Increased intracranial pressure (b) Increased venous return (c) Increased functional residual capacity (d) Cephalad displacement of the diaphragm

**Ans: (c) Increased functional residual capacity** — Trendelenburg **decreases** FRC and compliance because abdominal contents push the diaphragm cephalad → atelectasis. It increases venous return, CVP, **ICP and IOP**, and risks facial/laryngeal oedema and tube migration.

**Q14. Reverse Trendelenburg position is typically used for**

- (a) Abdominal hysterectomy (b) Thyroidectomy (c) Haemorrhoidectomy (d) Nephrectomy

**Ans: (b) Thyroidectomy** — Reverse Trendelenburg (head up) is used for **neck surgery — thyroidectomy, parathyroidectomy, scalene node biopsy — and laparoscopic cholecystectomy**. Trendelenburg is for pelvic surgery, lithotomy/jackknife for anorectal, lateral kidney for nephrectomy.

**Q15. The Kraske (jackknife) position is a modification of the**

- (a) Supine position (b) Prone position (c) Lateral position (d) Fowler's position

**Ans: (b) Prone position** — **Kraske/jackknife is a PRONE modification** exposing the sacrococcygeal and anorectal area. **Trendelenburg, reverse Trendelenburg, Fowler's and lithotomy are all modifications of SUPINE.**

**Q16. Both legs must be lowered slowly and simultaneously from the lithotomy position mainly to prevent**

- (a) Nerve traction (b) A sudden fall in blood pressure (c) Wound dehiscence (d) Deep vein thrombosis

**Ans: (b) A sudden fall in blood pressure** — Lowering the legs shifts blood into the lower limbs → **sudden reduction in venous return and a precipitous fall in blood pressure**. Two persons lower them together, slowly.



**Q17. In the lateral position the axillary (chest) roll should be placed**

(a) Directly in the axilla (b) Just caudal to the axilla under the chest wall (c) Under the iliac crest (d) Between the knees

**Ans: (b) Just caudal to the axilla under the chest wall** — It goes **caudal to (below) the axilla** to lift the chest wall off the **dependent brachial plexus and axillary vessels**. Placing it **in** the axilla causes the very injury it should prevent.

**C — Environment, Asepsis and SSI**

**Q18. A HEPA filter removes**

(a) 99.97 % of particles  $\geq 0.3 \mu\text{m}$  (b) 99.9 % of particles  $\geq 1 \mu\text{m}$  (c) 95 % of particles  $\geq 0.5 \mu\text{m}$  (d) 99.999 % of particles  $\geq 0.1 \mu\text{m}$

**Ans: (a) 99.97 % of particles  $\geq 0.3 \mu\text{m}$  — HEPA = 99.97 % of particles  $0.3 \mu\text{m}$  and larger.** The 99.999 % at  $0.1 \mu\text{m}$  figure is an **ULPA** filter, used in surgical-smoke evacuators.

**Q19. The operating room is maintained at which pressure relative to the corridor?**

(a) Negative (b) Positive (c) Equal (d) Alternating

**Ans: (b) Positive — Positive**, so air flows **out** of the OR. **Negative pressure with  $\geq 12$  air changes/hour** is used for **airborne isolation — tuberculosis, measles, varicella**.

**Q20. The recommended relative humidity in an operating room is**

(a) Below 20 % (b) 20–60 % (c) 70–80 % (d) Above 80 %

**Ans: (b) 20–60 % — 20–60 %** (classically 30–60 %). **Too low  $\rightarrow$  static electricity and spark risk; too high  $\rightarrow$  microbial growth and condensation.** Temperature **20–23 °C** with  **$\geq 15$  air changes/hour**.

**Q21. The single most effective measure for preventing healthcare-associated infection is**

(a) Prophylactic antibiotics (b) Hand hygiene (c) Ultraviolet irradiation (d) Routine fumigation

**Ans: (b) Hand hygiene — Hand hygiene with an antimicrobial agent** — stated explicitly in your PDF as the greatest single factor in preventing nosocomial infection. Routine formaldehyde fumigation is obsolete and UV is only an adjunct.

**Q22. A draped instrument table is considered sterile**

(a) To 30 cm below the table top (b) Only at and above table-top level (c) Down to the floor (d) Only in the central third

**Ans: (b) Only at and above table-top level — Anything below table-top level is unsterile and is never brought back up.** Related: the wrapper edge (**1 inch / 2.5 cm**) is unsterile, the **gown back** is unsterile, and **moisture contaminates by strike-through**.

**Q23. An elective cholecystectomy without spillage is classified as**

(a) Clean (b) Clean-contaminated (c) Contaminated (d) Dirty

**Ans: (b) Clean-contaminated — Entry into the biliary/GI tract under controlled conditions = clean-contaminated (Class II), SSI 3–10 %.** Clean **< 2 %**, contaminated **15–20 %**, dirty/infected **30–40 %**.

**Q24. The commonest organism causing surgical site infection is**

(a) Escherichia coli (b) Pseudomonas aeruginosa (c) Staphylococcus aureus (d) Clostridium perfringens

**Ans: (c) Staphylococcus aureus — Staphylococcus aureus**, then coagulase-negative staphylococci, Enterococcus and E. coli. The commonest **source** of contamination is the **patient's own endogenous flora**.

**Q25. Alcohol-based hand rub is ineffective, and soap and water must be used, against**

(a) MRSA (b) Clostridioides difficile (c) Influenza virus (d) Pseudomonas aeruginosa

**Ans: (b) Clostridioides difficile — Alcohol is not sporicidal, so C. difficile spores need mechanical removal with soap and water** (plus contact precautions and hypochlorite for surfaces). Soap and water is also mandatory for **visibly soiled hands**.

**Q26. Artificial fingernails are prohibited in the OR chiefly because they**

- (a) Tear surgical gloves (b) Harbour Gram-negative bacteria and fungi (c) Interfere with pulse oximetry (d) Absorb chlorhexidine

**Ans: (b) Harbour Gram-negative bacteria and fungi** — AORN prohibits artificial nails and chipped polish because the **subungual space harbours Gram-negative bacilli and fungi** and cannot be decontaminated. (Polish does also affect pulse oximetry, but infection control is the reason asked.)

**Q27. The correct sequence for REMOVING (doffing) personal protective equipment is**

- (a) Gown → gloves → mask → goggles (b) Gloves → goggles → gown → mask (c) Mask → gown → gloves → goggles (d) Goggles → gloves → mask → gown

**Ans: (b) Gloves → goggles → gown → mask** — **Doffing: gloves → goggles/face shield → gown → mask/respirator (outside the room) → hand hygiene.** The most contaminated item comes off first and the respirator last. **Donning is the reverse: gown → mask → goggles → gloves.**

**D — Electro Surgery and Equipment**

**Q28. Electrosurgical units operate in which frequency range?**

- (a) 50–60 Hz (b) 1–10 kHz (c) 300 kHz – 3 MHz (d) 10–100 MHz

**Ans: (c) 300 kHz – 3 MHz — Radiofrequency, 300 kHz – 3 MHz.** Above about **100 kHz** the current no longer stimulates nerve or muscle (**the Faradic effect is avoided**), which is why RF current cuts and coagulates without electrocuting.

**Q29. Which statement about bipolar electrosurgery is TRUE?**

- (a) It requires a dispersive return pad (b) Current passes through the whole patient (c) No dispersive pad is required (d) It cannot be used in patients with pacemakers

**Ans: (c) No dispersive pad is required** — Both electrodes are the **two tines of the forceps**, so current passes only between them. **No pad is needed**, lateral spread is minimal, and it is the **safest mode with a pacemaker**.

**Q30. The coagulation waveform of an ESU is**

- (a) Continuous low-voltage current (b) Interrupted high-voltage current (c) Direct current (d) Continuous current at 50 Hz

**Ans: (b) Interrupted high-voltage current** — **Coag = interrupted, high-voltage, low duty cycle (~6 %); cut = continuous, low-voltage, high-current (100 % duty cycle).** Coag's high voltage explains its greater risk of **capacitive coupling and insulation breakdown**.

**Q31. Capacitive coupling in laparoscopic electrosurgery is most likely when**

- (a) Bipolar forceps are used (b) A metal instrument passes through a plastic (hybrid) trocar (c) The pad is on the thigh (d) Cut mode is used at low power

**Ans: (b) A metal instrument passes through a plastic (hybrid) trocar** — **A hybrid metal-plastic trocar** lets current be induced through **intact insulation** into surrounding tissue. Use all-metal or all-plastic systems, lowest power, cut rather than coag, short activations and **active-electrode monitoring**.

**Q32. A pin-hole break in the insulation of a laparoscopic electrode classically causes**

- (a) An immediately visible spark (b) A return-electrode burn (c) Delayed bowel perforation several days later (d) Failure of the generator to activate

**Ans: (c) Delayed bowel perforation several days later** — **Insulation failure** discharges current **outside the surgeon's field of view** → a full-thickness bowel burn perforating **3–10 days post-operatively** with peritonitis. Inspect insulation before every use.

### Q33. A return-electrode contact quality monitoring (REM) system prevents

- (a) Alternate-site burns from metal contact (b) Pad-site burns from inadequate pad contact (c) Capacitive coupling (d) Surgical smoke exposure

**Ans: (b) Pad-site burns from inadequate pad contact — A split pad** whose halves are monitored for impedance; the generator **shuts down when contact becomes inadequate**, virtually eliminating **return-electrode (pad-site) burns**.

### Q34. The surgeon repeatedly asks for higher ESU power. The correct action is to

- (a) Increase the power as requested (b) Switch to bipolar (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty (d) Reposition the ECG electrodes

**Ans: (c) Stop, check all connections and the dispersive pad, and replace the unit if faulty —** Repeated requests indicate a **fault — poor pad contact, a loose connection, a frayed cord or a failing generator**. Check the circuit; if no cause is found, **turn off and unplug the unit, label it, get another, and document the occurrence and serial numbers**.

### Q35. Which statement about surgical smoke is correct?

- (a) A standard surgical mask is adequate protection (b) Ablating 1 g of tissue is roughly equivalent to 3–6 unfiltered cigarettes (c) It is sterile and harmless (d) A HEPA rather than an ULPA filter is required in the evacuator

**Ans: (b) Ablating 1 g of tissue is roughly equivalent to 3–6 unfiltered cigarettes —** Smoke contains **> 150 chemicals** plus viable cells and viral DNA (**documented HPV transmission to surgeons**). Control = **smoke evacuator with an ULPA filter held within 1–2 inches** plus a **high-filtration mask or N95** — a standard mask is **not** sufficient.

### Q36. For an average adult, maximum recommended tourniquet time and pressure for the LOWER limb are

- (a) 30 min at 200 mmHg (b) 60 min at 250 mmHg (c) 90–120 min at 300–350 mmHg (d) 3 hours at 400 mmHg

**Ans: (c) 90–120 min at 300–350 mmHg — Lower limb 300–350 mmHg for ≤ 90–120 min; upper limb 250–300 mmHg for ≤ 60 min. Deflate for 10–15 min** if longer, and record site, pressure and both inflation and deflation times.

## E — Sterilization and Disinfection

### Q37. Standard gravity-displacement autoclave conditions are

- (a) 100 °C at atmospheric pressure for 30 min (b) 121 °C at 15 psi for 15–20 min (c) 134 °C at 15 psi for 30 min (d) 160 °C for 2 hours

**Ans: (b) 121 °C at 15 psi for 15–20 min — 121 °C (250 °F) at 15 psi for 15–20 minutes, or 134 °C at 30 psi for 3–3.5 minutes.** 160 °C for 2 h is a **hot air oven (dry heat)** cycle.

### Q38. Flash (immediate-use) steam sterilization of an unwrapped metal instrument is carried out at

- (a) 121 °C for 20 min (b) 132 °C for 3 min (c) 132 °C for 30 min (d) 160 °C for 1 hour

**Ans: (b) 132 °C for 3 min — 132 °C (270 °F) for 3 minutes** unwrapped non-porous metal; **10 minutes** for porous, lumened or complex items. IUSS is **only for an urgently needed single item** — never implants, sets, or convenience.

### Q39. The Bowie-Dick test detects

- (a) Inadequate air removal in a prevacuum sterilizer (b) Adequate ethylene oxide concentration (c) Spore-killing efficacy (d) Steam superheating only

**Ans: (a) Inadequate air removal in a prevacuum sterilizer — A Class 2 chemical indicator** run **daily in an EMPTY prevacuum chamber at 134 °C for 3.5 min** to test **air removal / steam penetration**. It does **not** prove sterility — only a **biological indicator** does.

**Q40. The biological indicator for steam sterilization is**

- (a) *Bacillus atrophaeus* (b) *Geobacillus stearothermophilus* (c) *Bacillus pumilus* (d) *Clostridium sporogenes*

**Ans: (b) *Geobacillus stearothermophilus* — *Geobacillus stearothermophilus*** (thermophile, incubated 55–60 °C) for **steam and hydrogen peroxide plasma**; ***Bacillus atrophaeus*** for **dry heat and ethylene oxide**; ***Bacillus pumilus*** for **gamma radiation**.

**Q41. Dry heat sterilization in a hot air oven requires**

- (a) 121 °C for 15 min (b) 134 °C for 3 min (c) 160 °C for 2 hours (d) 180 °C for 2 hours

**Ans: (c) 160 °C for 2 hours — 160 °C for 2 h** (or 170 °C for 1 h, 180 °C for 30 min). It is preferred for **glassware, oils, powders, greases and sharp instruments — it does not blunt or corrode them —** but not for rubber, plastics or linen.

**Q42. Which item CANNOT be processed in a hydrogen peroxide gas plasma (STERRAD) sterilizer?**

- (a) A fibre-optic cable (b) A stainless-steel instrument (c) Cotton gauze and paper drapes (d) A microsurgical instrument

**Ans: (c) Cotton gauze and paper drapes — Cellulose — paper, cotton, linen, gauze, dressings — absorbs the sterilant**, as do liquids, powders and long narrow lumens. Advantages: **45–50 °C, ≈ 1 h cycle, and NO aeration** because by-products are only water and oxygen.

**Q43. Sterilization with 2 % activated glutaraldehyde requires immersion for**

- (a) 10 minutes (b) 30 minutes (c) 3 hours (d) 10 hours

**Ans: (d) 10 hours — 10 hours for sterilization; high-level disinfection in 20–45 minutes** (your PDF quotes 10 min — quote the textbook if the question comes from it). Rinse in **sterile distilled water** and **test potency with a strip** before each use.

**Q44. After use, peracetic acid decomposes into**

- (a) Formaldehyde and water (b) Acetic acid, water and oxygen (c) Chlorine and water (d) Ethylene glycol

**Ans: (b) Acetic acid, water and oxygen — Acetic acid (vinegar), water and oxygen —** so it is environmentally safe and can go down an ordinary drain. It also works **in the presence of organic soil** and its low surface tension reaches lumens.

**Q45. Under the Spaulding classification, a flexible gastrointestinal endoscope is**

- (a) Critical — needs sterilization (b) Semi-critical — needs high-level disinfection (c) Non-critical — needs low-level disinfection (d) Non-critical — needs only cleaning

**Ans: (b) Semi-critical — needs high-level disinfection —** Endoscopes contact **intact mucous membranes → semi-critical → high-level disinfection**. **Critical** items enter sterile tissue or the vascular system (sterilization); **non-critical** items touch only intact skin.

**Q46. The internationally accepted Sterility Assurance Level is**

- (a)  $10^{-3}$  (b)  $10^{-4}$  (c)  $10^{-6}$  (d)  $10^{-12}$

**Ans: (c)  $10^{-6}$  — SAL =  $10^{-6}$  —** no more than one viable organism per million sterilized items. This is the quantitative definition of 'sterile'.

**Q47. Which is the MOST resistant to sterilization and disinfection?**

- (a) HIV (b) *Mycobacterium tuberculosis* (c) Bacterial spores (d) Prions

**Ans: (d) Prions —** Decreasing resistance: **prions > bacterial spores > coccidia > mycobacteria > small non-lipid viruses > fungi > vegetative bacteria > lipid-enveloped viruses (HIV, HBV — the easiest to kill)**. Prions need **1 N NaOH plus 134 °C for 18 min**.

**Q48. The shelf-life of a sterilized wrapped pack is determined by**

- (a) A fixed 7-day limit (b) A fixed 30-day limit (c) Event-related sterility (d) The class of chemical indicator used

**Ans: (c) Event-related sterility — Event-related sterility —** the pack is sterile until an event compromises it (wet, torn, punctured, seal broken, dropped, over-handled). The old "30 days" rule is obsolete.

**Q49. Gamma radiation sterilization uses a cobalt-60 source at a dose of**

- (a) 2.5 kGy (b) 25 kGy (c) 250 kGy (d) 2500 kGy

**Ans: (b) 25 kGy — 25 kGy (2.5 Mrad)** — a **cold, industrial** method for pre-packed **single-use disposables** (syringes, gloves, catheters, sutures, blades, heart valves), **not used inside hospitals** because of cost and shielding.

**Q50. Membrane filtration of a heat-labile solution uses a pore size of**

- (a) 0.45 µm (b) 0.22 µm (c) 1.2 µm (d) 5 µm

**Ans: (b) 0.22 µm — 0.22 µm** retains bacteria and is used for sera, antibiotic and vaccine solutions and media. Note it **removes but does not kill**, and **does not retain viruses or mycoplasma** (0.1 µm is needed).

**F — Emergencies and Disasters**

**Q51. Malignant hyperthermia is triggered by**

- (a) Propofol and nitrous oxide (b) Volatile inhalational agents and succinylcholine (c) Non-depolarising muscle relaxants (d) Local anaesthetics

**Ans: (b) Volatile inhalational agents and succinylcholine** — Triggers are **all volatile agents and succinylcholine**. **Safe: propofol, thiopentone, benzodiazepines, opioids, nitrous oxide, non-depolarising relaxants and all local anaesthetics**. The defect is in the **RYR1 gene on chromosome 19**.

**Q52. The initial dose of dantrolene in malignant hyperthermia is**

- (a) 0.5 mg/kg (b) 1 mg/kg (c) 2.5 mg/kg (d) 10 mg/kg immediately

**Ans: (c) 2.5 mg/kg — 2.5 mg/kg IV bolus, repeated every 5-10 min up to 10 mg/kg** (occasionally 30). Simultaneously stop the triggers, **hyperventilate with 100 % O<sub>2</sub> at ≥ 10 L/min**, cool actively (**stop at 38 °C**), treat hyperkalaemia, and keep urine output **> 1-2 mL/kg/h**.

**Q53. The earliest sign of malignant hyperthermia under general anaesthesia is**

- (a) Hyperthermia (b) An unexplained rise in end-tidal CO<sub>2</sub> (c) Myoglobinuria (d) Hypotension

**Ans: (b) An unexplained rise in end-tidal CO<sub>2</sub> — Rising EtCO<sub>2</sub>** with tachycardia and tachypnoea; **masseter spasm** after succinylcholine is the classic clue. **Hyperthermia is a LATE sign** — waiting for it costs lives.

**Q54. The adult intramuscular dose of adrenaline in anaphylaxis is**

- (a) 0.1 mg of 1:10 000 (b) 0.5 mg of 1:1000 into the antero-lateral thigh (c) 1 mg of 1:1000 subcutaneously (d) 5 mg nebulised

**Ans: (b) 0.5 mg of 1:1000 into the antero-lateral thigh — 0.5 mg IM (0.5 mL of 1:1000) into the antero-lateral thigh, repeated every 5 min; 50 µg IV boluses** if monitored and experienced. Then **crystalloid 20 mL/kg**, chlorpheniramine, hydrocortisone and salbutamol.

**Q55. Latex anaphylaxis under anaesthesia characteristically**

- (a) Never causes hypotension (b) Is typically delayed 20-60 minutes after exposure (c) Occurs only in adults (d) Responds only to steroids

**Ans: (b) Is typically delayed 20-60 minutes after exposure** — Absorption across **mucosa or peritoneum** makes latex reactions **delayed 20-60 min** rather than immediate. Highest risk: **spina bifida, children with multiple operations, health-care workers**. Cross-reacting foods: **banana, avocado, chestnut, kiwi**.

**Q56. The extinguisher of choice for an electrical fire in the OR is**

- (a) Water (b) Foam (c) Carbon dioxide (Class C) (d) Class D dry powder

**Ans: (c) Carbon dioxide (Class C) — CO<sub>2</sub> or ABC dry chemical for Class C electrical fires — never water**. CO<sub>2</sub> is preferred around equipment because it leaves no residue. The commonest **ignition** source of OR fires is the **ESU (70-90 %)** and the commonest **fuel** is **alcohol-based prep**.

**Q57. In START triage, a casualty who is not breathing even after the airway is opened is tagged**

- (a) Red (b) Yellow (c) Green (d) Black

**Ans: (d) Black — Black = dead/expectant. Red** = immediate (RR > 30, absent radial pulse or capillary refill > 2 s, or not obeying commands); **Yellow** = delayed; **Green** = minor/walking. START assesses **RPM — Respiration, Perfusion, Mental status** in under a minute.

## **G — Radiation, Chemical Hazards and Waste**

**Q58. The SI unit of equivalent dose of radiation is the**

- (a) Gray (b) Sievert (c) Becquerel (d) Roentgen

**Ans: (b) Sievert — Sievert (Sv) — used for all dose limits and personnel monitoring (1 Sv = 100 rem).** The **Gray** is absorbed dose (1 Gy = 100 rad) and the **Becquerel** is activity.

**Q59. The annual occupational whole-body radiation dose limit is**

- (a) 1 mSv (b) 20 mSv averaged over 5 years (c) 50 mSv every year without restriction (d) 150 mSv

**Ans: (b) 20 mSv averaged over 5 years — 20 mSv/year averaged over 5 consecutive years (100 mSv/5 y), with a maximum of 50 mSv in any single year.** Public **1 mSv/y**; lens **20 mSv/y**; extremities **500 mSv/y**; declared pregnancy **≤ 1 mSv to the fetus (0.5 mSv/month)**.

**Q60. If the distance from a radiation source is doubled, exposure is reduced to**

- (a) One half (b) One quarter (c) One eighth (d) One sixteenth

**Ans: (b) One quarter — Inverse square law — intensity  $\propto 1/d^2$ , so doubling gives one quarter and tripling gives one ninth. Stand  $\geq 2$  m (6 ft) away or leave the room. The three principles are **time, distance and shielding**, under **ALARA**.**

**Q61. A personal radiation dosimeter should be worn**

- (a) Inside the lead apron at waist level (b) Outside the lead apron at chest/collar level (c) On the wrist (d) On the X-ray machine

**Ans: (b) Outside the lead apron at chest/collar level — Outside the apron at chest or collar level, to record dose to the unshielded neck, head and lens. A pregnant worker wears a second badge under the apron at waist level. TLD is the commonest type.**

**Q62. Radiation-induced carcinogenesis is an example of a**

- (a) Deterministic effect with a threshold (b) Stochastic effect with no threshold (c) Acute deterministic effect (d) Non-radiation effect

**Ans: (b) Stochastic effect with no threshold — Stochastic effects (cancer, heritable mutation) have no threshold — probability rises with dose. Deterministic effects (erythema, epilation, cataract, sterility, marrow suppression) have a threshold and their severity rises with dose.**

**Q63. The OSHA permissible exposure limit (8-h TWA) for ethylene oxide is**

- (a) 0.05 ppm (b) 1 ppm (c) 25 ppm (d) 100 ppm

**Ans: (b) 1 ppm — EO = 1 ppm (excursion 5 ppm/15 min). Others: glutaraldehyde ceiling 0.05 ppm, formaldehyde 0.75 ppm, nitrous oxide 25 ppm (NIOSH), halogenated volatiles 2 ppm, methyl methacrylate 100 ppm.**

**Q64. Under the Biomedical Waste Management Rules 2016, a used syringe with a fixed needle is discarded into a**

- (a) Yellow bag (b) Red bag (c) White translucent puncture-proof container (d) Blue container

**Ans: (c) White translucent puncture-proof container — Sharps, including syringes with fixed needles, go in the WHITE translucent puncture-proof container → autoclaved and shredded, then encapsulated or sent to a foundry. Red = recyclable plastics (syringes without needles); Yellow = anatomical, soiled, expired drugs, cytotoxics; Blue = glass and metallic implants.**

## **H — Post-Operative Patient Care**

**Q65. Which of the following is NOT a component of the modified Aldrete score?**

- (a) Respiration (b) Circulation (c) Pain (d) Oxygen saturation

**Ans: (c) Pain** — The modified Aldrete score assesses **Activity, Respiration, Circulation, Consciousness and Oxygen saturation (A-R-C-C-O)**, each 0-2, maximum 10, **discharge  $\geq 9$** . Pain and nausea belong to the **PADSS**, used for discharge home after day surgery.

**Q66. The original Aldrete score used which parameter that the modified score replaced with pulse oximetry?**

- (a) Temperature (b) Skin colour (c) Urine output (d) Bowel sounds

**Ans: (b) Skin colour** — The 1970 score used **skin colour**; the modified score substituted **SpO<sub>2</sub>**, which is objective and more sensitive.

**Q67. The commonest cause of upper airway obstruction in the immediate post-anaesthesia period is**

- (a) Laryngospasm (b) The tongue falling back against the posterior pharynx (c) Bronchospasm (d) Vocal cord paralysis

**Ans: (b) The tongue falling back against the posterior pharynx** — Loss of pharyngeal tone in the sedated supine patient lets the **tongue fall backwards** — recognised by **snoring and see-saw (paradoxical) chest movement**. Treat with **jaw thrust or head tilt-chin lift, lateral position, suction and an airway**.

**Q68. The commonest post-operative pulmonary complication is**

- (a) Pneumonia (b) Atelectasis (c) Pulmonary embolism (d) Pneumothorax

**Ans: (b) Atelectasis** — **Atelectasis** — typically low-grade fever on **POD 1-2** with reduced basal breath sounds. Treatment is **deep breathing, incentive spirometry, splinted coughing, early ambulation and adequate analgesia** — not antibiotics.

**Q69. Fever on the second post-operative day is most likely due to**

- (a) Wound infection (b) Atelectasis (c) Deep vein thrombosis (d) Urinary tract infection

**Ans: (b) Atelectasis** — Follow the W's in time order: **Wind (atelectasis, POD 1-2) → Water (UTI, POD 3-5) → Walking (DVT, POD 4-6) → Wound (SSI, POD 5-7) → Wonder drugs (> POD 7)**. Fever in the **first 24 h is usually not infective**.

**Q70. Fever on the sixth post-operative day with wound erythema and purulent discharge indicates**

- (a) Atelectasis (b) Drug fever (c) Surgical site infection (d) Malignant hyperthermia

**Ans: (c) Surgical site infection** — **SSI classically declares itself on POD 5-7**. Management: **open, drain and culture the wound, debride and pack for healing by secondary intention**, with culture-guided antibiotics.

**Q71. Wound dehiscence most commonly occurs on which post-operative day, and its herald sign is**

- (a) POD 1-2; bright red bleeding (b) POD 5-10; serosanguineous 'salmon-pink' discharge (c) POD 14-21; purulent discharge (d) POD 30; a palpable lump

**Ans: (b) POD 5-10; serosanguineous 'salmon-pink' discharge** — Dehiscence peaks at **POD 5-10 (classically day 7-8)**, when the wound has only **5-10 %** of its final tensile strength. The warning sign is a **sudden serosanguineous (salmon-pink) discharge** with a palpable gap.



### Q72. The correct immediate action for post-operative wound evisceration is to

(a) Push the viscera back and apply a dry dressing (b) Cover the viscera with sterile gauze soaked in warm sterile saline, position low Fowler's with knees flexed, keep nil by mouth and call the surgeon (c) Sit the patient upright and give oral fluids (d) Apply a tight binder and discharge

**Ans: (b) Cover the viscera with sterile gauze soaked in warm sterile saline, position low Fowler's with knees flexed, keep nil by mouth and call the surgeon — Never replace the viscera and never use dry gauze.** Cover with **warm sterile saline-moistened gauze**, position supine/low Fowler's with knees flexed to reduce tension, keep **nil by mouth**, secure IV access, monitor for shock and **prepare for immediate theatre**.

### Q73. Sutures on the face are typically removed on day

(a) 3-5 (b) 7-10 (c) 10-14 (d) 14-21

**Ans: (a) 3-5 — Face and neck 3-5 days** (rich blood supply, cosmesis); scalp and trunk **7-10**; limbs **10-14**; joints, back, palms and soles **14 days**.

### Q74. Post-operative return of bowel function occurs in which order?

(a) Colon → stomach → small bowel (b) Small bowel → stomach → colon (c) Stomach → colon → small bowel (d) All simultaneously at 72 hours

**Ans: (b) Small bowel → stomach → colon — Small intestine ≈ 24 h → stomach 24-48 h → COLON 48-72 h (last to recover).** Resolution is judged by the **passage of flatus or stool**, not bowel sounds.

### Q75. Which is a CLOSED active suction drain?

(a) Penrose drain (b) Corrugated drain (c) Jackson-Pratt drain (d) Open sump drain

**Ans: (c) Jackson-Pratt drain — Jackson-Pratt (bulb) and Hemovac/Redivac (spring or high vacuum) are closed active systems with a lower infection risk; Penrose and corrugated drains are passive and open.** A fully re-expanded JP bulb means suction has been lost — re-compress it after emptying.

### Q76. According to the Apfel score, which is NOT a risk factor for PONV?

(a) Female sex (b) Non-smoking status (c) Smoking (d) Expected post-operative opioid use

**Ans: (c) Smoking —** The four Apfel factors are **female sex, non-smoker, previous PONV or motion sickness, and expected post-operative opioid use** — giving ≈ **10/20/40/60/80 %** risk for 0-4 factors. **Smoking is protective.**

### Q77. Secondary haemorrhage after surgery classically occurs on which day, and why?

(a) Day 0; slipped ligature (b) Within 24 h; clot dislodged as blood pressure rises (c) POD 7-14; infection eroding a vessel wall (d) POD 30; false aneurysm

**Ans: (c) POD 7-14; infection eroding a vessel wall — Primary = during surgery; reactionary = within 24 h (slipped ligature or clot dislodged as BP rises and vasoconstriction wears off); secondary = POD 7-14 from infection eroding the vessel wall.**

## 18. RAPID REVISION

### Mnemonics

| Mnemonic                                       | Stands for                                                                                                                                                                         |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2 - 4 - 6 - 8</b>                           | Fasting: clear fluids 2 h, breast milk 4 h, formula/light meal 6 h, fatty meal 8 h                                                                                                 |
| <b>Sign In - Time Out - Sign Out</b>           | The 3 phases of the WHO Surgical Safety Checklist                                                                                                                                  |
| <b>PADDING</b>                                 | Positioning: <b>P</b> ressure points padded, <b>A</b> lignment, <b>D</b> ignity, <b>D</b> evelopes ready, <b>I</b> V lines protected, <b>N</b> erves free, <b>G</b> entle movement |
| <b>Critical - Semi-critical - Non-critical</b> | Spaulding: sterilize - high-level disinfect - low-level disinfect                                                                                                                  |



| Mnemonic                                                            | Stands for                                                                                                                                         |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Geo = Steam ; Atro = Air &amp; EO</b>                            | <b>Geobacillus</b> stearothermophilus → <b>steam</b> and plasma; Bacillus <b>atrophaeus</b> → dry heat (air) and <b>Ethylene oxide</b>             |
| <b>R - A - C - E</b>                                                | Fire: <b>R</b> escue, <b>A</b> larm, <b>C</b> onfine, <b>E</b> xtinguish/Evacuate                                                                  |
| <b>P - A - S - S</b>                                                | Extinguisher: <b>P</b> ull, <b>A</b> im at the base, <b>S</b> queeze, <b>S</b> weep                                                                |
| <b>R - P - M</b>                                                    | START triage: <b>R</b> espiration, <b>P</b> erfusion, <b>M</b> ental status                                                                        |
| <b>4 H's &amp; 4 T's</b>                                            | Reversible causes of arrest: Hypoxia, Hypovolaemia, Hypo/Hyperkalaemia-metabolic, Hypothermia; Tension pneumothorax, Tamponade, Toxins, Thrombosis |
| <b>4 D's</b>                                                        | Negligence: <b>D</b> uty, <b>D</b> ereliction, <b>D</b> irect causation, <b>D</b> amage                                                            |
| <b>A - R - C - C - O</b>                                            | Aldrete: <b>A</b> ctivity, <b>R</b> espiration, <b>C</b> irculation, <b>C</b> onsciousness, <b>O</b> xygen saturation — max 10, discharge ≥ 9      |
| <b>Wind - Water - Walking - Wound - Wonder drugs</b>                | Post-op fever by day: 1-2 → 3-5 → 4-6 → 5-7 → > 7                                                                                                  |
| <b>Time - Distance - Shielding</b>                                  | Radiation protection, under ALARA                                                                                                                  |
| <b>Yellow burns • Red recycles • White is sharp • Blue is glass</b> | Biomedical waste colours                                                                                                                           |
| <b>'Only what is documented is a legal fact'</b>                    | Perioperative documentation                                                                                                                        |
| <b>'When in doubt, throw it out'</b>                                | Doubtful sterility = unsterile                                                                                                                     |
| <b>'The treatment of pus is drainage'</b>                           | Surgical site infection                                                                                                                            |

## The 25 statements most likely to be a one-mark question

1. The **circulator** keeps the legal record of the count and is the **patient's advocate**.
2. **At least two identifiers**, and the patient **states** their own name.
3. The site is marked by the **operating surgeon** with an **indelible marker**, with the patient participating.
4. The **time out** is **after induction, before skin incision**; the WHO checklist has **3 phases**.
5. **Hearing is the last sense to be lost** — silence during induction.
6. The **safety strap across the mid-thighs** is the key measure against a fall from the table.
7. **Ulnar nerve** — commonest peri-operative nerve injury; **common peroneal** in lithotomy (**foot drop**).
8. **Trendelenburg raises ICP and IOP and lowers FRC**; **Kraske/jackknife is a PRONE modification**.
9. **Never move an anaesthetised patient without the anaesthetist's permission**.
10. The patient's **skin must never touch metal** when the ESU is in use (alternate-site burn), and the **pencil stays in its holster**.
11. ESU works at **300 kHz-3 MHz**; **bipolar needs no pad** and is safest with a pacemaker; **coag = interrupted high voltage**.
12. Autoclave **121 °C / 15 psi / 15-20 min**; hot air oven **160 °C / 2 h**; flash **132 °C / 3 min**.
13. **Bowie-Dick tests air removal**; the **biological indicator is the only proof of sterilizing efficacy**; **SAL = 10<sup>-6</sup>**.
14. **EO requires aeration**; **plasma cannot be used for cellulose**; **glutaraldehyde 10 h sterilizes**.

15. **HEPA = 99.97 % at 0.3  $\mu\text{m}$** ; the OR is **positive** pressure with  **$\geq 15$  air changes/hour**.
16. **Hand hygiene is the single most effective measure** against healthcare-associated infection.
17. **A draped table is sterile only at table-top level**; the wrapper edge (**1 inch**) is unsterile;  
**moisture = contamination**.
18. **X-ray-detectable sponges are never used as dressings; nothing leaves the OR until the final count is correct**.
19. **MH: volatile agents + succinylcholine; dantrolene 2.5 mg/kg; earliest sign is rising  $\text{EtCO}_2$** .
20. **Anaphylaxis: adrenaline 0.5 mg IM 1:1000; latex reactions are delayed 20-60 min**.
21. The commonest **ignition** source of OR fires is the **ESU**; the commonest **fuel** is **alcohol prep**;  **$\text{CO}_2$  for electrical fires**.
22. **Aldrete  $\geq 9$  to leave PACU; PADSS  $\geq 9$  to go home; pain is in PADSS, not Aldrete**.
23. Commonest immediate PACU emergency = **airway obstruction from the tongue**; commonest post-op pulmonary complication = **atelectasis**.
24. **Dehiscence POD 5-10 with serosanguineous discharge; evisceration  $\rightarrow$  warm sterile saline-soaked gauze and call the surgeon**.
25. **Radiation: 20 mSv/year; doubling the distance quarters exposure; the dosimeter goes OUTSIDE the apron**.

*End of exam-focused notes • Revise the PDF for the Chapter 1-3 core text, and these 18 sections for every gap.*



